Arithmetic landscapes III:
deformed measures, random sequences, and the coset identity

(subtitle provisional until the Part-III assembly is complete)

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Abstract

For a finite set A of odd positive integers write $F_A(z) = \prod_{a\in A}(1 + z^a)$ for the generating polynomial of the subset sums, so that $r_A(n) = \#\{S \subseteq A : \sigma(S) = n\}$ is its n-th coefficient. Everything below concerns $|F_A|$ on the unit circle, where $|F_A(e^{i\theta})| = 2^{|A|}\prod_{a\in A}|\cos(a\theta/2)|$, and the per-element quantity $X(t) = -\log|\cos \pi t|$ that governs it.

Our first result is an identity for X alone, elementary and exact:

$$\frac{1}{v}\sum_{k=0}^{v-1} X\Big(t + \frac{k}{v}\Big) \;=\; \Big(1 - \frac{1}{v}\Big)\log 2 \;+\; \frac{1}{v}X(vt + \tau_v) \qquad \text{for every } v \geq 1 \text{ and every } t,$$

with $\tau_v \in \{0, \frac{1}{2}\}$ explicit. Averaging X over a coset of index v returns the same function at v times the frequency, plus a constant. We prove it twice, by the multiplication formula and by Fourier, and it is what the minor-arc argument of this Part runs on: the step that normally costs a quantitative equidistribution estimate costs nothing.

Parts I and II [3, 4], and the first half of this Part, fix two things at once — the set A, and the uniform measure on its subsets. The rest of this Part moves each. Replacing the uniform measure by a Bernoulli(q) product measure deforms the gap series of Part I into an explicit family $\Gamma^{(q)}$, and we prove that the fair coin minimises it for every set, with equality only at $q = \frac{1}{2}$; the argument is termwise and is machine-checked in Lean 4. We derive the corresponding transfer function $\Phi^{(q)}$ and show that the bias splits the transfer-function strand's [5] symmetric pair of strata in the ratio $q : (1 - q)$, so that the biased landscape responds to a shift of target at first order; the resulting slope is a point prediction with no free parameter, and it is confirmed against the natural alternative to within 0.4%. Replacing the primes by a random odd sequence, we prove that the extremal modulus of the minor arcs moves from 6 to 4, so that the peak height available to a random odd sequence is $1/\sqrt{2}$ where the primes have $\sqrt{3}/2$ — the primes are the harder case. We then prove that the geometric *rate* follows: almost surely the minor-arc maximum of $F_A^{1/b}$ tends to $1/\sqrt{2}$. The proof carries no ineffective constant and needs no equidistribution input, because the coarse-graining identity replaces the usual discrepancy estimate by an equality. Two questions inherited from the transfer-function strand are also settled or sharpened. Every claim is tagged with its status, and the numerical work is reproducible from logged scripts.

1 Introduction

Parts I and II and the transfer-function strand fixed two things at once: the sequence A, and the measure on its subsets. Every statement of the trilogy is about a specific A under the

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uniform measure. This paper moves each of them, one at a time, and the two moves turn out to be worth making for different reasons.

Bias. The classification theorem of Part I [3] says which elements lie on which side of a target. It says nothing about how configurations are weighted, so the entire combinatorial layer survives verbatim when the uniform measure is replaced by the Bernoulli(q) product measure, and the gap series deforms into an explicit one-parameter family $\Gamma^{(q)}$ (§8). The answer has a shape worth putting in the introduction: **the fair coin gives the least rugged landscape**, for every sequence, by a termwise convexity argument that is machine-checked (Theorem 8.5, Remark 8.4). The transfer function deforms too, and here the deformation is not cosmetic: the transfer-function strand's cosh splits into two unequal exponentials with offsets in the ratio $q : (1-q)$, so the biased landscape responds to a shift of target at *first* order rather than second (§9). Rigidity off the fair coin is a tilt, not a plateau.

Coarse-graining. §3 is one lemma and it is the technical spine of the paper. Averaging the energy $X(t) = -\log|\cos \pi t|$ over the coset $\{t + k/v\}$ returns X at v times the frequency, plus $(1 - 1/v)\log 2$, exactly (Lemma 3.1, proved by the multiplication formula and again by Fourier). Since $X \geq 0$, the rational points are therefore the *minima* of the coset average, not merely convenient sample points (Corollary 3.2) — so the step of a minor-arc argument that normally costs a quantitative equidistribution estimate costs nothing here. Read as physics it is a block-spin step with an exact fixed point; we say so once, in Remark 3.4, and deduce nothing from the reading.

Randomness. For a random odd sequence the minor-arc input that cost Part II [4] fifty rounds is replaced by a concentration bound, and §10 carries *no ineffective constant at all* — the first such statement in the series. What we did not expect is that the random case is also *easier*: the extremal modulus moves from 6 to 4 (Theorem 10.5), so the peak height drops from $\sqrt{3}/2$ to $1/\sqrt{2}$, and the geometric rate improves in step (Theorem 10.8). The modulus moves for a one-line arithmetic reason — a set of primes contains at most one element $\equiv 3 \pmod 6$, a random odd set contains a third of them — and the consequence is that Part II's $\sqrt{3}/2$ is a property of the primes, not a limitation of the method. *The primes are the hard case.*

A methodological remark that this paper earned twice. Deformations are not only a way to generalise; they are a way to *see*. **A deformation that breaks a symmetry makes measurable everything the symmetry was hiding.** Constructing $\Gamma^{(q)}$ and $\Phi^{(q)}$ audited the transfer-function strand for free, twice. First, the c_d fork: asking how the omitted factor behaves as k grows, rather than how large it is at one k, closed a question the transfer-function strand had left open (§11). Second, and more sharply, the sign of λ: the untilted Φ of §5 is an *even* function of λ, so no observable in that paper depends on the sign, and a definition that carried the wrong one survived three papers and an external referee. Splitting the cosh made the sign observable in a single measurement. We record the principle because it is a design rule, not an anecdote: when a quantity cannot be checked, deform the problem until it can.

Verification, and the genre. Every numbered claim carries a status tag: proved, Lean-verified, experimentally confirmed with the range stated, or conjectured. The scripts that produce every number in this paper write logs beside themselves and are part of the artefact. This is now a recognisable genre — machine-assisted mathematics shipped with formal certificates, of which *Ten advances in mathematics and theoretical computer science* [1] and its accompanying Lean 4 repository are the visible precedent. The direction differs: that work attacks named open problems, whereas the present series builds a small theory from its definitions upward, so the certificates here anchor *definitions and structural lemmas* rather than headline resolutions. The verification ethos is the same one, and we take it as read rather than arguing for it.

1.1 The transfer-function strand

Parts I and II [3, 4] of this series establish that for the odd primes $\mathrm{lm}/r \to \Gamma$ *at the centre* of the landscape. Whether Γ is an invariant of the landscape or an accident of its most symmetric point is the question this paper answers, and the answer turns out to be stronger than rigidity alone: the whole ratio is a function of one local observable. The argument and what each part rests on:

- Parts I and II establish $\mathrm{lm}/r \to \Gamma$ *at the centre* $n = \lfloor T/2 \rfloor$: Part I reduces it to a flatness hypothesis and Part II discharges that hypothesis unconditionally. The obvious question — whether Γ is a property of the landscape or of its centre — is what this paper answers.
- The combinatorial layer (classification, exact stratification, sandwich) was proved *for every target* in Part I and is formally verified; nothing there needs redoing. [Lean-verified]
- The minor-arc estimates of Part II [4] bound $|G_B(\theta)|$ as a function of θ only. They do not refer to the extraction point m, which enters $r_B(m) = \frac{1}{2\pi}\int F_B(\theta)e^{-im\theta}d\theta$ only through a phase. They therefore transfer to off-centre targets unchanged. [quoted from Part II]
- What is new is the middle, §5.2: an additive bridge (Lemma 5.8) carrying the minor-arc estimates from the uniform measure to the exponential tilt, the minor-arc input in the form the bridge needs (Lemma 5.10), and a tilted local limit theorem uniform over the extraction window (Proposition 5.20). The odd orders in λ cancel by the reflection symmetry of the landscape, which is where cosh enters.
- One consequence of quoting rather than reproving deserves saying at the outset: *the only ineffective constant in the trilogy is one and the same Siegel–Walfisz constant, and it enters each paper at exactly one point* — here, part (b) of Lemma 5.10. See §14.

2 Notation, and what is inherited

The definitions are those of Part I [3], written so that nothing assumes $n = T/2$. Fix throughout an increasing sequence $A = (a_1, \dots, a_k)$ of odd positive integers, $T = \sigma(A)$, $S_2 = \sum a_i^2$, $V = S_2/4$, and a target n. Write $\rho = n/T$ and $x = \frac{1}{2} - \rho$. For $d \geq 1$ put

$$N_d = \#\{i : a_i \leq 2d\}, \qquad \sigma_d = \sum_{a_i \leq 2d} a_i, \qquad s_d = \sum_{a_i \leq 2d} a_i^2, \qquad \delta_d = d + \tfrac{\sigma_d}{2},$$

and let $B_d = \{a_i : a_i > 2d\}$ be the layer-d ensemble, of variance $V_d = (S_2 - s_d)/4 = V - s_d/4$.

Definition 2.1 (local logarithmic slope)**.** $\lambda(n)$ is the logarithmic derivative of the representation count at the target,

$$\lambda(n) \;=\; +\frac{d}{dm}\log r_A(m)\Big|_{m=n} \;=\; \tfrac{1}{4}\log\frac{r_A(n+2)}{r_A(n-2)} + O(\cdot),$$

so $\lambda > 0$ below the centre, where r_A is increasing.

Remark 2.2 (on the sign, and why it took the biased deformation to matter). An earlier version of this definition carried the opposite sign, which made it disagree with Remark 2.3 (where $\lambda \approx (T/2-n)/V > 0$ below the centre) and with Remark 2.5 (where $s/\lambda \to 1$ with $s > 0$ below the centre). Nothing in this paper changed when the sign was corrected, and nothing could have: Φ is an *even* function of λ, so the sign of λ is not observable anywhere in the transfer function. It becomes observable only when the cosh is split into two unequal exponentials, which is what happens under a biased measure; the discrepancy was found there and is recorded here. The convention above is the one that makes $\lambda \approx s$.

Remark 2.3 (why the slope and not the offset). The Gaussian approximation $\lambda \approx (T/2-n)/V$ is what one writes down first, and it is wrong at the 10% level for the purposes of this paper. Substituting the measured λ for the Gaussian one takes the drift of the fitted coefficient across the range $x \in [0.06, 0.30]$ from 15–18% down to 0.03–0.07%, and it dissolves a non-integer exponent that an earlier draft of this programme had reported as a finding: the exponent was an artefact of the wrong variable. This is not a cosmetic choice: every statement below is false at the stated precision if $(T/2-n)/V$ is substituted for λ.

Definition 2.4 (the tilt)**.** P_s is the product measure on subsets under which a_i is present with probability $p_a = (1+e^{sa})^{-1}$, and $s = s(n) > 0$ is chosen so that $\sum_a a\, p_a = n$; thus P_s is the exponential tilt of the uniform measure whose mean sits at the target.

Remark 2.5 (the tilt and the slope agree to first order). s and λ are the same quantity to leading order but not exactly: s makes the tilted *mean* equal n, whereas λ is the slope of $\log r_A$ *at* n, and the two differ by the mean–mode gap. Measured on three profiles,

$$\frac{s(n)}{\lambda(n)} \;=\; 1 + \frac{c_A}{k} + O(k^{-2}), \qquad c_A \approx 1.85 \text{ (odds)},\; 2.03 \text{ (primes)},\; 2.87 \text{ (squares)},$$

stable in x and clean in k over $k = 100, \ldots, 220$. [experimentally confirmed; `tilt_r088`] The statements below are formulated in λ, and *not* in s: substituting s multiplies the residual of Table 1 by three, exactly as the λ^2 leading behaviour of Φ predicts. [experimentally confirmed; `sres_r088`]

3 An exact coarse-graining identity

Parts I and II and the transfer-function strand borrowed the vocabulary of statistical physics — landscape, energy, transfer function, tilt — and used it as a source of the right questions. This section is the first place where the borrowing runs the other way and produces an exact recursion rather than an analogy. Everything later in the paper that concerns minor arcs (§10.1, §10.2) is a corollary of it.

Throughout write

$$X(t) \;=\; -\log|\cos \pi t| \;\in [0,\infty],$$

the per-element energy of Parts I and II and the transfer-function strand at phase t.

Lemma 3.1 (coset identity)**.** *For every integer $v \geq 1$ and every real t,*

$$\frac{1}{v}\sum_{k=0}^{v-1} X\Big(t+\frac{k}{v}\Big) \;=\; \Big(1-\frac{1}{v}\Big)\log 2 \;+\; \frac{1}{v}\,X(vt+\tau_v), \qquad \tau_v = \begin{cases} 0, & v \text{ odd},\\ \frac{1}{2}, & v \text{ even}. \end{cases}$$

Equivalently $\prod_{k<v} |\cos\pi(t+k/v)| = 2^{\,1-v}\,|\cos\pi(vt+\tau_v)|$.

First proof (multiplication formula). Write $2\cos\pi z = 2\sin\pi(z+\frac{1}{2})$ and apply $\prod_{k<v} 2\sin\pi(y+\frac{k}{v}) = 2\sin(v\pi y)$ — the factorisation of $x^v - 1$ read on the unit circle — with $y = t + \frac{1}{2}$. The product of the $2|\cos|$ is $2|\sin\pi(vt+\frac{v}{2})|$, which is $2|\cos\pi vt|$ for v odd and $2|\sin\pi vt| = 2|\cos\pi(vt+\frac{1}{2})|$ for v even. Divide by 2^v and take logarithms. □

Second proof (Fourier). From $-\log|2\sin\pi u| = \sum_{n\geq 1}\cos(2\pi nu)/n$ and $u = t+\frac{1}{2}$,

$$X(t) = \log 2 + \sum_{n\geq 1}\frac{(-1)^n\cos(2\pi nt)}{n}.$$

Averaging over the coset annihilates every harmonic but those with $v \mid n$, since $v^{-1}\sum_{k<v}\cos(2\pi n(t+k/v))$ is $\cos(2\pi nt)$ when $v \mid n$ and 0 otherwise. Writing $n = vp$ and using $(-1)^{vp} = (-1)^p$ for v odd and 1 for v even, the surviving series is $v^{-1}(X(vt+\tau_v) - \log 2)$. □

Two proofs check the *statement*; one proof checks only the argument. The identity is verified in product form for $v \leq 60$ at 200 random shifts each, worst discrepancy $2.5\cdot 10^{-16}$, and the two Fourier partial sums agree to $8.8\cdot 10^{-14}$. [proved, twice; and checked numerically, `stab_r110`, `rate_r112`]

Corollary 3.2 (the rational points are the minima). $\displaystyle\frac{1}{v}\sum_{k<v} X(t+\tfrac{k}{v}) \;\geq\; \Big(1-\frac{1}{v}\Big)\log 2$ *for every t, with equality exactly when $vt+\tau_v \in \mathbb{Z}$.*

Proof. $X \geq 0$. □

Remark 3.3 (what is machine-checked in Lemma 3.1, and what is not). The identity is *multiplicative in v*, and that is what has been formalised. Writing $k = i + aj$ and inducting on b splits a product over $\{k < ab\}$ into a double product with no bijection and no analysis — `prod_range_mul_index`, stated for an arbitrary commutative monoid. On top of it, `cosetId_one` and `cosetId_two` are the base cases ($v = 2$ is the double-angle formula and needs nothing else), and `cosetId_mul` carries the identity from a and b to ab, including the offset bookkeeping $a\tau_b + \tau_a \equiv \tau_{ab} \pmod 1$ in all four parity cases, via `abs_cos_add_int_mul_pi`. *What is not formalised is the base case at an odd prime*, which is the only place a product over roots of unity is needed, and which Mathlib does not supply. So the machine-checked statement is: *the identity holds at 1 and 2 and is closed under products*, hence holds for every v built from the primes at which it is established by hand. The reduction was checked numerically before it was formalised, including a deliberately wrong τ as a control. [Lean-verified (the reduction; not the odd-prime base case); `Pnp/Theory/CosetProd.lean`, `coset_mult_r116`]

Remark 3.4 (what this says, and in which language). The identity has two readings and they have different statuses; the paper keeps them apart.

As arithmetic (proved) it says that averaging X over a coset of index v returns X *itself, at v times the frequency*, plus a constant — so the family $\{X(v\,\cdot)\}$ is closed under coarse-graining, exactly, with no error term. That is what Corollary 3.2 turns into a uniform lower bound, and it is why §10.2 needs no equidistribution input: the question "how far can the mean fall when the rational grid is shifted?" has the answer *it cannot fall at all.*

As physics (an interpretation, and nothing is deduced from it) this is a block-spin step. Coarse-graining the phase lattice by a factor v maps the energy to a rescaled copy of itself; the map has an exact fixed-point structure rather than an approximate one, and the constant $(1-1/v)\log 2$ is the free energy released per coarse-graining step, additive over successive steps because τ composes. Parts I and II and the transfer-function strand treated the landscape picture as a source of questions; this is the first identity in the series that the picture predicts and the arithmetic confirms. §13 says what we would want to do with it and does not do it here.

4 Rigidity

Theorem 4.1 (rigidity of the gap series). *[conditional on Problem 12.1 of §12 and on nothing else; that problem is the whole of what is missing, and it is shared with Proposition 5.20 and Theorem 5.1 — see the consumer list there. The missing ingredient is named rather than left to be located. The analytic ingredients are proved in §5.2; what is* not *written out is the Edgeworth expansion in region R1 of Proposition 5.20 — a classical local-limit computation under a product measure, given here only to the level of its cumulant budget, with the explicit constants missing. Until those constants are written, the statement below is conditional on that computation and on nothing else.] Let $P = P_k$ be the first k odd primes and let $\rho \in (0,1)$ be fixed. Then*

$$\frac{\mathrm{lm}_P(\lfloor \rho T \rfloor)}{r_P(\lfloor \rho T \rfloor)} \xrightarrow[k\to\infty]{} \Gamma(P),$$

uniformly for ρ in compact subsets of $(0,1)$. For a general profile A the same holds under hypothesis (H) *of §7.*

$\Gamma(A)$ is therefore an invariant of the landscape and not of its centre. Theorem 4.1 is the case $\lambda \to 0$ of Theorem 5.1: at fixed ρ one has $\lambda(n) \asymp x/N \to 0$, and Φ is continuous at 0 with $\Phi(0) = \Gamma$, so

$$\Phi(\lambda) - \Gamma \;=\; O\Big(\lambda^2 \sum_{d\ge 1} 2^{1-N_d}\delta_d^2\Big) \;\longrightarrow\; 0,$$

the sum being finite precisely by (H). The uniformity in ρ needs no new estimate: every bound in §5.2 is uniform once $t_{\min}(\rho)$ is bounded below, and $t_{\min}$ is continuous and positive on $(0,1)$, hence bounded below on compact subsets (Remark 5.19).

[experimentally confirmed] The evidence the theorem is modelled on. With $\mathrm{ratio}(\rho,k) = [\mathrm{lm}/r]/\Gamma(P_k)$, the spread over ρ contracts monotonically:

k	40	52	64	76	88	100
$\max_\rho - \min_\rho$	0.01051	0.00283	0.00257	0.00162	0.00113	0.00083
$\max_\rho \lvert\mathrm{ratio} - 1\rvert$	0.00921	0.00276	0.00244	0.00156	0.00109	0.00081

over $\rho \in [0.20, 0.50]$, by exact integer computation. Three guards are in force throughout this paper's computations and are worth stating once: every reported point satisfies $r \ge 10^4$, so that integer granularity cannot masquerade as signal; the dynamic programme was checked against brute-force enumeration over all 2^k subsets at small k; and the exact identity $\mathrm{lm}(n) = \mathrm{lm}(T-n)$ is verified at every measured target, which tests the stratification and the arithmetic together.

5 The transfer function

Theorem 5.1 (transfer). *[conditional on Problem 12.1 of §12 and on nothing else; that problem is the whole of what is missing, and it is shared with Proposition 5.20 and Theorem 5.1 — see the consumer list there. The missing ingredient is named rather than left to be located. The analytic ingredients are proved in §5.2; what is* not *written out is the Edgeworth expansion in region R1 of Proposition 5.20 — a classical local-limit computation under a product measure, given here only to the level of its cumulant budget, with the explicit constants missing. Until those constants are written, the statement below is conditional on that computation and on nothing else.] Let*

$A = P_k$ be the first k odd primes, or a general profile satisfying (H)*, and let $n = \lfloor \rho T \rfloor$ with $\rho \in (0,1)$ fixed. Then*

$$\frac{\mathrm{lm}_A(n)}{r_A(n)} = \Phi_k(\lambda(n)) + E_k, \qquad \Phi_k(\lambda) = 1 + \sum_{d\geq 1} 2^{1-N_d}\, e^{-\lambda^2 s_d/8} \cosh(\lambda\delta_d), \tag{1}$$

with $E_k \to 0$ as $k \to \infty$, uniformly for ρ in compact subsets of $(0,1)$.

Remark 5.2 (what the theorem does and does not claim about E_k)*.* The assertion is $E_k \to 0$ in absolute terms, nothing sharper. The measured rate $E_k \asymp c_A/k$ of Remark 5.7 is *not* part of the statement; it is quoted as [experimentally confirmed, $k \leq 220$, three profiles] and stated as an open problem in §14. Distinguishing the two matters here more than usual, because the numerical agreement is far better than what is proved.

Three features of (1) deserve to be stated before any proof.

Remark 5.3 (the ratio sees the target only through one number)*.* Φ depends on A alone. The target enters only through $\lambda(n)$. This is a stronger statement than rigidity: rigidity says the limit does not move, (1) says how it fails to move.

Remark 5.4 ($\lambda = 0$ recovers Part I [3])*.* The layer weight $e^{-\lambda^2 s_d/8}$ is 1 at $\lambda = 0$, so $\Phi(0) = 1 + \sum_{d\geq 1} 2^{1-N_d}$, which is the window identity $W_D(A) = \Gamma(A) + (2D+1)2^{-k}$ of Part I — already formally verified. Numerically, $\Phi(0) - \Gamma(A)$ is 0, $-4.7 \cdot 10^{-14}$ and $-2.7 \cdot 10^{-15}$ for the three sequences of Table 1.

Remark 5.5 (each layer is a smaller ensemble)*.* The factor $e^{-\lambda^2 s_d/8}$ is not a fudge: it is what remains when the Gaussian normalisations of B_d and of A fail to cancel, because $V_d = V - s_d/4$. Expanding to second order gives the coefficient of λ^2 as $\delta_d^2 - s_d/4$ rather than δ_d^2. In the notation of §2, layer d is the ensemble B_d with the small elements removed, so its variance is $V_d = V - s_d/4$ rather than V; the weight $e^{-\lambda^2 s_d/8}$ is exactly the price of that difference, and Remark 5.21 derives it from the ratio of partition functions rather than fitting it. The correction is termwise smaller than the term it corrects, $s_d/4 \leq d\sigma_d/2 < \delta_d^2$, so every term of the series stays positive and the correction never overshoots.

5.1 Verification

[experimentally confirmed] Table 1 evaluates (1) at the measured $\lambda(n)$ against the exactly computed lm/r.

$x = \frac{1}{2} - \rho$	0.06	0.08	0.10	0.15	0.20	0.25	0.30
odds, $k = 220$	0.84%	0.85%	0.85%	0.86%	0.88%	0.92%	0.97%
primes, $k = 220$	0.91%	0.92%	0.92%	0.94%	0.96%	0.99%	1.05%
$a_i \asymp i^{3/2}$, $k = 220$	1.05%	1.06%	1.06%	1.08%	1.10%	1.14%	1.20%
squares, $k = 170$	1.66%	1.67%	1.68%	1.70%	1.74%	1.80%	1.89%

Table 1: Residual $|\,\text{measured} - \Phi(\lambda)\,|$ **as a fraction of the centred signal** $\mathrm{lm}/r - [\mathrm{lm}/r]_{n=T/2}$. Absolute residuals are 10^{-5} to 10^{-6}; quoting them as significant figures of the ratio would be meaningless, since the signal itself lives in the fifth and sixth digit. The third row is the only ensemble in this paper that was *constructed to test a prediction* rather than chosen because it was natural: $\alpha = 3/2$ was built after Remark 5.7 had been fitted on the other three, and its predicted constant $16/7$ lies between theirs.

Remark 5.6 (why the residual is quoted this way)*.* Against the *uncentred* signal the same residuals read 2% to 300%, the large values occurring precisely where the signal is smallest.

Neither number is wrong; the centred one is the one the theorem is about, because $\Phi(0) = \Gamma$ is an identity and the finite-k offset $[\mathrm{lm}/r]_{n=T/2} - \Gamma$ is a separate quantity of size 10^{-6}. We keep this remark because an earlier draft of this programme reported "agreement to five or six significant figures" for a model that was 12% wrong on the signal: the figures were real and the claim was not, because the signal itself lives in those digits.

Remark 5.7 (the residual obeys a $1/k$ law). Table 1 is one column of a k-scan. Writing $R_A(k)$ for the residual in the sense of the caption, the measurement over $k = 60, \ldots, 220$ gives

$$k\,R_A(k) \;\longrightarrow\; \begin{cases} 1.85\% & \text{odds } (k = 60, 100, 140, 180, 220:\ \ 2.29, 2.00, 1.91, 1.87, 1.85), \\ 2.01\% & \text{primes } (k = 100, \ldots, 220:\ \ 2.38, 2.06, 2.03, 2.01), \\ 2.82\% & \text{squares } (k = 130, \ldots, 190:\ \ 3.78, 2.85, 2.83, 2.82). \end{cases}$$

Two things follow. First, the residual is $O(1/k)$ and therefore vanishes, which is the form Theorem 5.1 asserts; the table's 1% is a finite-k number and not a floor. Second, the squares row is the loosest of the three because c_A is larger, not because the decay is different: squares obey the same law with a constant 1.52 times that of the odd numbers. The constants c_A coincide numerically with those of Remark 2.5 — the same mean–mode gap seen twice — but substituting s for λ does not remove the residual, so the coincidence is a shared order of magnitude and not a shared cause. [experimentally confirmed; `tilt_r088`, `sres_r088`]

What they share is a driver. The saddle-point representation gives the exact identity

$$s(n) - |\lambda(n)| \;=\; \frac{K'''(s)}{2K''(s)^2}, \qquad K'' = \sum a^2 p_a(1-p_a), \quad K''' = \sum a^3 p_a(1-p_a)(1-2p_a),$$

tested pointwise to a relative error $0.54/k$ [experimentally confirmed; `saddle_r092`], and for small tilt $K'''/(2K''^2) \approx sS_4/S_2^2$, so that $c'_A = k(s/|\lambda| - 1) \to kS_4/S_2^2$. For $a_i \asymp i^\alpha$ this is $(2\alpha+1)^2/(4\alpha+1)$. The measured c_A of this remark agrees with c'_A to within 1% on four profiles:

	α	kS_4/S_2^2	c'_A	c_A	c_A/c'_A
odd numbers	1	1.8000	1.8525	1.8674	1.008
$a_i \asymp i^{3/2}$	3/2	2.2794	2.3494	2.3318	0.993
squares	2	2.7714	2.8615	2.8377	0.992
primes	—	1.9731	2.0251	2.0253	1.000

at $k = 220$. The two halves of this have different statuses and the difference matters:

- $c'_A \to (2\alpha+1)^2/(4\alpha+1)$ is *derived* — the identity above is exact, its sign is forced by $\Lambda''' < 0$ for $\rho < \frac{1}{2}$, and the small-tilt step is a Taylor expansion — and [experimentally confirmed on five profiles, one of them out of sample].
- $c_A = c'_A(1 + o(1))$ is [experimentally confirmed: four profiles, ratios 0.992–1.008, $k = 220$]. The exact asymptotic equality, and the mechanism behind it, remain a **conjecture**. What is established is a shared driver, S_4/S_2^2, and *not* a causal link: substituting s for λ makes the residual three times worse, not smaller.

Carried to $k = 300$, the identification tightens rather than drifting, and it does so at a measurable rate. The ratio c_A/c'_A reads

k	120	180	220	260	300
odd numbers	1.0589	1.0244	1.0151	1.0098	1.0064
primes	1.0318	1.0084	1.0039	1.0014	1.0001
$a_i \asymp i^{3/2}$	1.0113	1.0015	0.9995	0.9984	—
squares	1.0782	1.0039	0.9985	—	—

so $|c_A/c'_A - 1|$ falls by a factor 9.2 across a factor 2.5 in k for the odd numbers, an exponent of 2.4 — the gap between the two constants closes *faster* than the $1/k$ law they both sit on. [experimentally confirmed, four profiles, $k \le 300$; e8_r103] This is a sharper statement of the same conjecture, not a proof of it: a rate of empirical convergence is evidence about a limit, and the mechanism is still missing.

5.2 The bridge to Part II

Both theorems reduce to the same five parts. **P1** is the exact combinatorial stratification, proved for every target in Part I [Lean-verified]; **P3** is a local limit theorem for the tilted ensemble of Definition 2.4; **P4** is the assembly, in which the odd orders in λ cancel by the reflection symmetry $\operatorname{lm}(n) = \operatorname{lm}(T-n)$ [Lean-verified]; **P5** is the error budget, which converges exactly under (H). Only **P2** is new: the minor-arc estimates of Part II [4] are proved for the *uniform* measure, and P3 needs them for the tilted one.

Write $\widetilde{G}_B(\theta) = \prod_{a\in B}(1 - p_a + p_a e(a\theta))$ for the tilted generating function and $S_B(\theta) = \sum_{a\in B} e(a\theta)$ for the linear exponential sum.

Lemma 5.8 (the bridge)**.** *Put $t_a = 4p_a(1-p_a)$ and $t_{\min} = \min_{a\in B} t_a$. For every θ,*

$$|\widetilde{G}_B(\theta)|^2 = \prod_{a\in B}\Big(1 - t_a \sin^2(\pi a\theta)\Big) \;\le\; \exp\Big(-t_{\min}\sum_{a\in B}\sin^2(\pi a\theta)\Big) \;=\; \exp\Big(-\tfrac{t_{\min}}{2}\big(b - \operatorname{Re} S_B(\theta)\big)\Big).$$

Consequently, if $\operatorname{Re} S_B(\theta) \le \eta b$ then $|\widetilde{G}_B(\theta)| \le \exp(-t_{\min}(1-\eta)b/4)$.

Proof. The identity $|1 - p + pe(\phi)|^2 = 1 - 4p(1-p)\sin^2(\pi\phi)$ is a computation; $t_a \ge t_{\min}$ and $1 - u \le e^{-u}$ give the middle step; and $\sum_a \sin^2(\pi a\theta) = \frac{1}{2}(b - \operatorname{Re} S_B(\theta))$ is the double-angle formula summed. ☐

Remark 5.9 (why the bound is additive, and a bound that is not available). It is tempting to transfer multiplicatively, as $|\widetilde{G}_B(\theta)| \le |G_B(\theta)|^\kappa$ with G_B the untilted product, since that would carry every exponent of Part II across at once. **No such $\kappa > 0$ exists.** At any θ with $a\theta \in \frac{1}{2} + \mathbb{Z}$ for some $a \in B$ the untilted factor $\cos(\pi a\theta)$ vanishes, so the right-hand side is 0, while the tilted factor equals $\sqrt{1 - t_a} > 0$. Equivalently $\log(1 - ty)/\log(1-y) \to 0$ as $y \to 1$. The underlying inequality $1 - ty \le (1-y)^t$ is false for every $t \in (0,1)$ — $s \mapsto (1-y)^s$ is convex and therefore lies *below* its chord $1 - sy$, with deficit exactly $-(1-t)$ at $y = 1$. Lemma 5.8 avoids the issue by never dividing by a quantity that vanishes. [refuted numerically on a 40 575-point grid and structurally; kappa_r088]

Lemma 5.10 (the minor-arc input)**.** *[(a) proved, Lemma 5.11; (b) proved, Lemma 5.12; (c) quoted from Part II] Let $A = P_k$ and $N = a_k$. Then*

$$\operatorname{Re} S_A(\theta) \;\le\; \Big(\tfrac{1}{2} + o(1)\Big) b \qquad \textit{uniformly for } \tfrac{1}{2N} \le \theta \le 1 - \tfrac{1}{2N}.$$

(a) $q \in \{1, 2\}$. *Lemma 5.11 below.*

(b) $3 \le q \le (\log N)^{A_0}$. *Lemma 5.12 below.*

(c) $q > (\log N)^{A_0}$. *This is the case $n = 1$ of Step 2 of* `prop:deepminor` *in Part II, which gives* $|S_A(\theta)| \ll N(\log N)^{3-A_0/2} + N^{4/5}(\log N)^3$, *i.e.* $\operatorname{Re} S_A/b \ll (\log N)^{4-A_0/2} = o(1)$ *for* $A_0 > 8$. *[quoted from Part II, verbatim]*

Lemma 5.11 (the two degenerate denominators). *Let $q \in \{1, 2\}$, $\theta = a/q + \beta$ with $\|\beta\| \geq 1/(2N)$. Then $\operatorname{Re} S_A(\theta) \leq \frac{1}{2}b$ for all large N.*

Proof. Write $S_A(\theta) = \sum_{p \leq N} e(pa/q)e(p\beta)$. For $q = 1$ the first factor is 1; for $q = 2$ it is $(-1)^p = -1$ on every odd prime, so $S_A = -\sum_{p \leq N} e(p\beta)$ and it suffices in both cases to bound $|\sum_{p \leq N} e(p\beta)|$. Partial summation against $\pi(t) = \operatorname{li}(t) + O(te^{-c\sqrt{\log t}})$ gives

$$\sum_{p \leq N} e(p\beta) = \int_2^N e(t\beta)\frac{dt}{\log t} + O\big(Ne^{-c\sqrt{\log N}}(1 + |\beta|N)\big),$$

and integrating by parts once more, $|\int_2^N e(t\beta)\, dt/\log t| \ll 1/(|\beta| \log N)$. With $\|\beta\| \geq 1/(2N)$ this is $\ll N/\log N \asymp b$ with an absolute implied constant, and the constant is $\leq \frac{1}{2}$ once $N\|\beta\|$ exceeds an absolute threshold; below that threshold, i.e. for $\|\beta\| \asymp 1/N$, the integral is $(1 + o(1))b \cdot \widehat{\chi}(\beta N)$ with $\widehat{\chi}$ the Fourier transform of the indicator of $[0, 1]$, whose real part is $\leq \frac{1}{2}$ off the principal arc. For $q = 2$ the sign is reversed, so $\operatorname{Re} S_A \leq 0$ near $\theta = \frac{1}{2}$ — the helpful direction. No Siegel–Walfisz is used. □

Lemma 5.12 (the small-q ring). *Let $3 \leq q \leq (\log N)^{A_0}$, $(a, q) = 1$, and $\theta = a/q + \beta$ with $|\beta| \leq 1/(q\tau)$, $\tau = N(\log N)^{-A_0}$. Then*

$$S_A(\theta) \;=\; \frac{\mu(q)}{\varphi(q)}\, W(\beta) \;+\; O\big(Ne^{-c\sqrt{\log N}}\big), \qquad W(\beta) = \int_2^N e(t\beta)\,\frac{dt}{\log t},$$

and $|W(\beta)| \leq \operatorname{li}(N) = (1 + o(1))b$. Consequently $\operatorname{Re} S_A(\theta) \leq (\frac{1}{2} + o(1))b$.

Proof. Sort the primes $p \leq N$ by their residue class. The $\omega(q) = O(\log q)$ primes dividing q contribute $O(\log q) = o(b)$, so

$$S_A(\theta) \;=\; \sum_{\substack{r \bmod q \\ (r,q)=1}} e\Big(\frac{ra}{q}\Big) \sum_{\substack{p \leq N \\ p \equiv r\,(q)}} e(p\beta) \;+\; O(\log q).$$

By Siegel–Walfisz, $\pi(t; q, r) = \operatorname{li}(t)/\varphi(q) + O(te^{-c\sqrt{\log t}})$ uniformly for $q \leq (\log N)^{A_0}$; partial summation against $e(t\beta)$ turns this into

$$\sum_{\substack{p \leq N \\ p \equiv r\,(q)}} e(p\beta) \;=\; \frac{W(\beta)}{\varphi(q)} \;+\; O\big(Ne^{-c\sqrt{\log N}}(1 + |\beta|N)\big),$$

and $|\beta|N \leq (\log N)^{A_0}/q$, so the error is $O(Ne^{-c\sqrt{\log N}}(\log N)^{A_0}) = O(Ne^{-c'\sqrt{\log N}}) = o(b)$: a fixed power of $\log N$ cannot compete with $e^{-c\sqrt{\log N}}$. Summing over r and using the Ramanujan sum $\sum_{(r,q)=1} e(ra/q) = c_q(a) = \mu(q)$ for $(a, q) = 1$ gives the display. Finally $\mu(q)/\varphi(q)$ is real and $\varphi(q) \geq 2$ for $q \geq 3$, so $\operatorname{Re} S_A(\theta) \leq (|\mu(q)|/\varphi(q))b(1+o(1)) \leq (\frac{1}{2}+o(1))b$; Lemma 5.17 says where the $\frac{1}{2}$ is attained. □

Remark 5.13 (where the ineffectivity is). The Siegel–Walfisz step in Lemma 5.12 is the only inexplicit estimate in §5.2 — Lemmas 5.8, 5.11, 5.17 and the Vaughan quotation in Lemma 5.10(c) are all effective — and it is the same constant that Part II isolates as C_1; see Remark 14.1.

Remark 5.14 (what would make C_1 effective). Two escapes are worth recording, because C_1 is the only inexplicit quantity in the trilogy. *Under GRH*, the Siegel–Walfisz input of Lemma 5.12 may be replaced by $\pi(t;q,r) = \mathrm{li}(t)/\varphi(q) + O(t^{1/2}\log t)$ with an absolute implied constant, uniformly for $q \le t^{1/2}$. The range Lemma 5.12 actually uses, $q \le (\log N)^{A_0}$, is far inside that, so on GRH the constant becomes effective and every constant in the trilogy is explicit. *Unconditionally*, the ineffectivity is confined to the possible existence of a Siegel zero for a single modulus $q \le (\log N)^{A_0}$: for every modulus with an explicit zero-free region the estimate is already effective. The inexplicit constant is therefore not a feature of the argument but a hostage to one hypothetical zero, and the range it is held over is only $(\log N)^{A_0}$, not N. Neither escape is used below; the theorems are stated unconditionally.

Remark 5.15 (the middle range, and why no constant enters). Inside $|\theta| < 1/(2N)$ Lemma 5.10 says nothing, and it does not have to: there $|a\theta| \le \frac{1}{2}$ for every $a \in A$, so $\sin^2(\pi a\theta) \ge 4a^2\theta^2$ and Lemma 5.8 gives the quadratic bound $|\widetilde{G}_B(\theta)| \le \exp(-4t_{\min}V_d\theta^2)$ directly. Cutting at the LCLT-natural radius $\theta_0 = (\log k)/\sqrt{K''_d}$ of §5.3, this is $\le \exp(-t_{\min}(\log k)^2)$ on the whole of $\theta_0 \le |\theta| \le 1/(2N)$ — superpolynomially below the density scale $(K''_d)^{-1/2}$, with no free constant anywhere. The crossover at $\theta = 1/(2N)$ survives as the boundary against Lemma 5.10, forced by $|a\theta| \le \frac{1}{2}$ rather than chosen; measured, the quadratic bound has already reached $e^{-0.01b}$ by $\theta N = 0.18$–0.42 across three profiles. [experimentally confirmed; `gap_r090`]

Remark 5.16 (what part (c) does *not* need). Part II must control every harmonic $n\theta$ with $n \le N_0$, and pays for it with a covering lemma and the parameter $Q_2 = (\log N)^{A_0 + A'_0}$. Here only $n = 1$ occurs, so the dichotomy in Lemma 5.10 is on the single denominator $q_1(\theta)$ and no covering lemma is needed. The quotation in (c) is therefore valid on a strictly larger set of θ than Part II's type (ii).

Lemma 5.17 (the extremiser of the additive estimate). $\max_{q\ge 3} \dfrac{\mu(q)}{\varphi(q)} = \dfrac{1}{2}$, *and the maximum is attained at $q = 6$ and at no other q.*

Proof. $\varphi(q) = 2$ holds exactly for $q \in \{3, 4, 6\}$, where μ takes the values -1, 0, $+1$; so among these the maximum is $\mu(6)/\varphi(6) = 1/2$. Every other $q \ge 3$ has $\varphi(q) \ge 4$ and hence $\mu(q)/\varphi(q) \le 1/4 < 1/2$. □

Remark 5.18 (the same $q = 6$, twice). By Lemma 5.10(b) the worst point of the additive estimate is therefore $\theta = 1/6$, with $\mathrm{Re}\, S_A/b \to \frac{1}{2} = \cos 60°$. Part II's multiplicative estimate has its worst point at the same q: $\sup_q M(q) = M(6) = \sqrt{3}/2 = \cos 30°$. **Same extremiser, same cause, two functionals** — both say that the odd primes lie in $\pm 1 \pmod 6$ and nowhere else, read once through $\prod|\cos|$ and once through $\sum \cos$. Measured on the odd primes below 10^7, $\mathrm{Re}\, S_A(1/6)/b = 0.499998$ against the predicted $1/2$, and $q = 6$ is the maximiser over $2 \le q \le 16$ at each of $N = 10^5, 10^6, 10^7$. [experimentally confirmed; `eta_r090`]

Remark 5.19 (the honest form of the tilt constant). $t_{\min} = t_{\min}(\rho)$ is positive for every fixed $\rho \in (0,1)$ and bounded below on compact subsets, which is all Lemma 5.8 needs. It is *not* governed by the approximation $sN \approx 6x$: that relation is the small-x asymptote only, and the measured ratio $sN/6x$ is 1.02–1.13 at $x = 0.10$ but 1.22–1.39 at $x = 0.30$. Measured values of $t_{\min}$: 0.91, 0.89, 0.91 at $x = 0.10$ and 0.36, 0.28, 0.34 at $x = 0.30$ for the odd numbers, squares and primes. [experimentally confirmed; `tilt_r088`, `kappa_r088`]

5.3 The tilted local limit theorem, and the assembly

Write $K''_d = \sum_{a\in B_d} a^2 p_a(1 - p_a)$ and $\mu_d(s) = \sum_{a\in B_d} a\, p_a$ for the variance and mean of B_d under P_s, and $K_d^{(j)}$ for the higher cumulants.

Proposition 5.20 (tilted LCLT, uniform over the window)**.** *[R2 and R3 proved (they quote Lemmas 5.8 and 5.10); R1 is Problem 12.1 of §12, the classical local limit computation with p_a-weights, written here to the level of its cumulant budget. Conditional on that problem and on nothing else.] For every layer $d \le D(k)$ and every m with $|m - \mu_d(s)| \le W(k)$,*

$$P_s[\textstyle\sum_{B_d} = m] = \frac{1}{\sqrt{2\pi K_d''}} \exp\Big(-\frac{(m-\mu_d)^2}{2K_d''}\Big)(1 + O(\varepsilon)), \quad \varepsilon = \frac{|K_d'''|}{(K_d'')^{3/2}} + \frac{K_d^{(4)}}{(K_d'')^2} + \frac{W(k)\,|K_d'''|}{(K_d'')^2},$$

uniformly in d and m.

Proof sketch. Invert $\widetilde{G}_{B_d}$ over $[-\pi, \pi]$ and split at $\theta_0 = (\log k)/\sqrt{K_d''}$ and at $1/(2N)$.

R1, $|\theta| \le \theta_0$. Expand $\log \widetilde{G}_{B_d}(\theta) = i\mu_d\theta - \frac{1}{2}K_d''\theta^2 - \frac{i}{6}K_d'''\theta^3 + \frac{1}{24}K_d^{(4)}\theta^4 + O(K_d^{(5)}\theta^5)$. Since $|K_d^{(j)}| \le \sum_a a^j$, on $|\theta| \le \theta_0$ every term beyond the quadratic is $o(1)$ and the standard Edgeworth argument gives the display with the stated ε; the third term contributes the $W|K'''|/(K'')^2$ piece over a window of width W.

R2, $\theta_0 \le |\theta| \le 1/(2N)$. Remark 5.15: the contribution is $\le \exp(-t_{\min}(\log k)^2)$, super-polynomially below $(K_d'')^{-1/2}$.

R3, $|\theta| \ge 1/(2N)$. Lemmas 5.8 and 5.10: $|\widetilde{G}_{B_d}| \le \exp(-t_{\min}(1-\eta)b/4)$, exponentially small, against measure $O(1)$.

Uniformity in d: B_d differs from A in $N_d \le N_{D(k)} = (\log k)^{O(1)}$ elements of size $\le 2D(k)$, so every cumulant shifts by a relative $O(s_D/S_2) \to 0$ and one ε serves all layers. The lattice is of step 1 and the parity weighting is quoted from Part II. □

Remark 5.21 (the μ_d-shift, and why the layer term is what it is). This is the step that produces Φ, so it is worth doing once in full. Since $\sum_{a\in A} a\,p_a = n$ and $p_a = \frac{1}{2} - \frac{sa}{4} + O((sa)^3)$ for the small elements,

$$\mu_d(s) \;=\; n - \sum_{a\le 2d} a\,p_a \;=\; n - \frac{\sigma_d}{2} + w + \dots, \qquad w := \frac{s\,s_d}{4},$$

so the two evaluation points of the stratification sit at

$$(n+d) - \mu_d = +\delta_d - w, \qquad (n-d-\sigma_d) - \mu_d = -\delta_d - w. \tag{2}$$

Feeding (2) into Proposition 5.20, undoing the tilt through $r_{B_d}(m) = P_s[\sum_{B_d} = m]\,Z_{B_d}(s)\,e^{sm}$, and using $sd + s\sigma_d/2 = s\delta_d$ to pair the two points:

$$r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d) = 2\,Z_{B_d} e^{sn - s\sigma_d/2}\,\frac{e^{-(\delta_d^2+w^2)/2K_d''}}{\sqrt{2\pi K_d''}}\cosh\Big(\delta_d\Big(s + \frac{w}{K_d''}\Big)\Big).$$

Dividing by $r_A(n) \sim Z_A e^{sn}/\sqrt{2\pi K''}$ and using $Z_{B_d}e^{-s\sigma_d/2}/Z_A = 2^{-N_d}\prod_{a\le 2d}\cosh(sa/2)^{-1} = 2^{-N_d}e^{-s^2 s_d/8}(1+\dots)$,

$$\frac{r_{B_d}(n+d) + r_{B_d}(n-d-\sigma_d)}{r_A(n)} = 2^{1-N_d}\,e^{-\lambda^2 s_d/8}\,\cosh(\lambda\delta_d)\cdot e^{-\delta_d^2/2K_d''}\,(1+O(\varepsilon)), \tag{3}$$

with $\lambda = s + w/K_d'' = s(1 + s_d/(4K_d''))$ — which is exactly the per-layer slope drift of $1 + O(s_d/S_2)$ measured earlier. **The first three factors of (3) are the general term of Φ in (1)**; the odd orders in λ cancel because the two points are $\pm\delta_d$ about the same centre, which is the reflection symmetry $\mathrm{lm}(n) = \mathrm{lm}(T-n)$ in analytic dress. [Lean-verified (the symmetry)]

Remark 5.22 (the one factor Φ omits). (3) carries a fourth factor, $c_d := e^{-\delta_d^2/2K''_d}$, that Φ does not. It is λ-*independent* and equal to $1 - O(\delta_d^2/V)$, so it belongs to E_k and not to the transfer function. Measured at $k = 180$: the ratio of the left side of (3) to Φ's term is c_d to six decimal places at $d = 1$ and $d = 3$ (1.000000 and 1.000003 for the odd numbers, 1.000000 and 1.000000 for the primes), and departs from 1 only at $d = D(k)$, where the layer weight is $2^{-29} \approx 2 \cdot 10^{-9}$. Carrying c_d inside Φ would move Table 1 from 0.842% to 0.814% (odd numbers, $k = 220$, $x = 0.06$), from 0.914% to 0.902% (primes) and from 1.052% to 1.048% ($\alpha = 3/2$): the factor is real, it moves the residual the right way, and it accounts for at most 3% of it. We therefore keep Φ as stated and count c_d in E_k. [experimentally confirmed; p3_r096, phig_r096]

That choice was made at one value of k, and the level of the correction is the wrong thing to decide it on; the trend is the right thing. Since $1 - c_d \asymp \delta_d^2/2V_d$ with $V \asymp k^{2\alpha+1}$, and the series is dominated by the small d where $\delta_d = O(1)$, the c_d correction is $O(k^{-(2\alpha+1)})$ while the residual is $O(1/k)$: the *share* of the residual that c_d carries should fall like $k^{-2\alpha}$. It does. Writing that share as $(R_\Phi - R_{\Phi c})/R_\Phi$ and fitting successive k:

$$1.98 \text{ (odd numbers, } \alpha = 1), \qquad 2.35 \text{ (primes)}, \qquad 2.99 \ (\alpha = \tfrac{3}{2}),$$

against the predicted 2α, i.e. 2.00, $\approx$2.4 for $a_i \asymp i \log i$, and 3.00; for the squares the prediction is k^{-4} and the correction is already at $1 - c_1 = 8.9 \cdot 10^{-10}$, so its share is numerical noise and changes sign — the prediction taken to its limit rather than a counterexample. Concretely the share for the odd numbers runs $10.74\%, 4.93\%, 3.33\%, 2.40\%, 1.81\%$ at $k = 120, 180, 220, 260, 300$. [experimentally confirmed, four profiles, $k \le 300$; e8_r103] Keeping Φ as printed is therefore not a convenience at the k we happened to measure: c_d's contribution is a vanishing fraction of a vanishing residual, so Φ is the right transfer function for every larger k and c_d belongs to E_k permanently. The fork is closed.

[experimentally confirmed; p3_r096] The verification behind Proposition 5.20 and (2), at $k = 180$, two profiles, $x = 0.10$ and $x = 0.30$, layers $d = 1, 3, D(k)$: the μ_d expression is exact to a relative $6 \cdot 10^{-5}$ or better; the two offsets of (2) to $7.6 \cdot 10^{-4}$ of δ_d or better; the R1 formula against the exact r_{B_d} over the window to $2.7 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$, in every case between 4% and 14% of the predicted ε; the parity oscillation of r_{B_d} over the window at most $3.7 \cdot 10^{-3}$; and ε itself grows by at most 44% from $d = 1$ to $d = D(k)$, so a single ε does serve all layers. With Lemmas 5.8 and 5.10 the tilted generating function decays like $\exp(-t_{\min}(1 - \eta)b/4)$ off the principal arc — exponentially in b, which is what P5 quotes. The tilted analogue of the $M(q)$ spectrum is lost, and is not needed: the rate enters Theorem 5.1 only through $E_k = o(1)$.

Remark 5.23 (why the constants cannot be imported). What Proposition 5.20 leaves open is not a mechanism but a list of explicit Edgeworth constants in region R1, so it is natural to ask whether they can be quoted rather than derived. The natural source is the Bernoulli-part-extraction literature, which produces local limit theorems with explicit universal constants for non-identically distributed lattice summands, and $\sigma(S) = \sum a_i X_i$ is a sum of independent lattice summands. We checked, and the answer is no, for two reasons; we record them because a reader will otherwise ask the same question.

Write, following Giuliano and Weber [2], $\vartheta_X = \sum_k \mathbb{P}\{X = k\} \wedge \mathbb{P}\{X = k + 1\}$ and $\Theta_n = \sum_j \vartheta_{X_j}$; their local limit theorem carries the error $C\Theta_n^{-1/2}(H_n + \Theta_n^{-1})$ plus a term $(\log \Theta_n/(\mathrm{Var} \cdot \Theta_n))^{1/2}$, with H_n a Berry–Esseen distance, and is useful exactly when $(\mathrm{Var}/\Theta_n)^{1/2}(H_n + \Theta_n^{-1}) \to 0$.

First, the hypothesis is void here. Our summands are $X_j = a_j X'_j$ with a_j odd and at least 3, so the law of X_j charges the two integers 0 and a_j and no two *adjacent* ones. Hence $\vartheta_{X_j} = 0$ exactly, for every j and every p, so $\Theta_n = 0$ and the bound reads $\le \infty$. Their own remark names this degenerate case.

Second, the obvious repair does not help. One can block the sequence and apply the method to block sums, whose laws do charge adjacent integers — but only once a block is large enough to have more subset sums than its span, and for blocks of *large* elements that takes a long time: $\{7, 9, 11\}$ has subset sums $\{0, 7, 9, 11, 16, 18, 20, 27\}$, no two adjacent. Measured, the smallest final block with $\vartheta > 0$ has size $7, 11, 15$ and more than 20 at $k = 16, 32, 64, 128$, growing like $\log k$, so blocking buys $\Theta_n \lesssim k/\log k$. And then the applicability condition fails outright: our weights grow, so $\operatorname{Var} \asymp S_2 \asymp k^3$ while $\Theta_n \lesssim k$ and $H_n \asymp k^{-1/2}$, giving $(\operatorname{Var}/\Theta_n)^{1/2}(H_n + \Theta_n^{-1}) \asymp k^{1/2}$. Measured, that quantity runs $48.9, 101.4, 222.5, 495.1$ for the odd numbers and $94.1, 195.1, 441.0, 1034.2$ for the primes at $k = 16, 32, 64, 128$ — increasing, not tending to zero. [experimentally confirmed; `bpart_r110`]

The obstruction is the same feature that makes the present local limit theorem worth proving: the summands carry weights $a_j \asymp j$, so the variance grows like the cube of the number of summands rather than linearly, and a method calibrated to the latter has nothing to give. *The Edgeworth constants of R1 have to be derived, not imported.*

6 Universality of the rate function

[experimentally confirmed, $k = 16..56$, three reference sequences] With $\hat{I}_A(\rho) = -k^{-1}\log(r_A(\lfloor \rho T \rfloor)/2^k)$ and $D_k = \hat{I}_P - \hat{I}_{\mathrm{ref}}$, the difference tends to 0 for each of three reference sequences: the odd numbers, the arithmetic progression $6n + 5$, and a random odd sequence of the same range. The sign is a property of the reference, not a universal fact — it is positive against the odd numbers, negative against the random sequence, and changes sign near $k = 48$ against the progression.

Remark 6.1 (the primes sit near the random sequence). At $k = 56$ the distance to a random odd sequence is 0.0016 against 0.0146 to the arithmetic progression — an order of magnitude. The reading we propose is that the rate function does not see the arithmetic of A beyond its counting function: the primes are close to a random sequence of the same density because, for this observable, that is what they are. It is the same statement the transfer function makes locally, since Φ depends on A only through (N_d, σ_d, s_d) — the small elements, counted. This is the most direct evidence we have for the reading that the main term is universal and the arithmetic of the sequence enters only through the correction.

7 A profile for which the hypothesis fails

The hypothesis Theorem 4.1 needs is the convergence, with room, of the series in (1) — qualitatively, that $N_d \to \infty$ fast enough. Precisely:

$$\textbf{(H)} \qquad \sup_k \sum_{d\geq 1} 2^{-N_A(d)}\delta_d^2 < \infty, \qquad \text{and} \qquad W(k) := \max\{\delta_d : 2^{-N_d}\delta_d^2 \geq k^{-1}\} = N^{o(1)}.$$

The first clause makes Φ converge and makes $\Phi(\lambda) - \Gamma = O(\lambda^2)$; the second is what the error budget needs, since the extraction window must be narrow enough for Proposition 5.20 to be uniform across it. For the odd primes $N_A(d) = \pi(2d) - 1 \gg d/\log d$, so both clauses hold, with $W(k) = O((\log k)^{2+\epsilon})$.

[experimentally confirmed] It is not vacuous. For the odd-ified cubes $a_i = 2\lfloor i^3/2 \rfloor + 1$:

profile	odds	primes	squares	cubes
N_d at $d = 10$ (any k)	10	7	4	**2**
$Q(0) = \Gamma^{-1}\sum 2^{-N_d}(\delta_d^2 - s_d/4)$	20.3	50.4	916	**381 904**

Only $\{1,9\}$ ever lie below 20, at any k, so 2^{-N_d} never damps the growing δ_d^2 and the series turns over only near $d \approx 10^4$. At $k=70$ the finite-size offset is still $1.75 \cdot 10^{-2}$ while the signal is $1.7 \cdot 10^{-11}$: nine orders of magnitude, so the profile is not merely slow but undecidable at any accessible size. The cubes are therefore not a defect of the theory but the reason (H) has to be there: without it, nothing in §5.2 or §5.3 fails, and yet the conclusion is out of reach.

7.1 Two different kinds of failure

The cubes fail *effectively* and not asymptotically, and the distinction is worth keeping. Computed directly, $\sum_d 2^{-N_d}\delta_d^2$ for the odd-ified cubes settles at $1.112 \cdot 10^7$ and does not grow with k (values at $k = 60, 100, 140, 180$: $1.112 \cdot 10^7$ from $k \approx 140$ on, against 69.5 for the odd numbers, 266.1 for $\alpha = 3/2$ and 6473 for the squares — each of them k-independent as well). [experimentally confirmed; `alpha_r092`] So (H) is not what the cubes violate; what they violate is any hope of seeing the asymptotics, the constant being four orders of magnitude larger than the odd numbers'.

A profile that fails (H) outright is available, and on the other side:

Proposition 7.1 (the $\alpha = 1$ threshold)**.** *Let $A = A_k$ consist of k distinct odd integers with $a_i \asymp c\, i^{\alpha}$.*

(i) If $\alpha < 1$ then (H) *fails: $\sum_{d\ge1} 2^{-N_d}\delta_d^2 \to \infty$.*

(ii) If $\alpha \ge 1$ then (H) *holds, with a bound depending on α and c alone.*

Proof. (i) The a_i are distinct odd integers, so $a_k \ge 2k-1$; with $a_k \asymp ck^{\alpha}$ this forces $c \gg k^{1-\alpha}$, so $a_1 \asymp c \to \infty$. For $d < a_1/2$ no a_i is $\le 2d$, hence $N_d = 0$, $\sigma_d = 0$, $\delta_d = d$, and

$$\sum_{d\ge1} 2^{-N_d}\delta_d^2 \;\ge\; \sum_{d<a_1/2} 2d^2 \;\asymp\; a_1^3 \;\asymp\; k^{3(1-\alpha)} \;\longrightarrow\; \infty.$$

(ii) For $\alpha \ge 1$ one has $a_1 = O(1)$ and $N_d = \#\{i : a_i \le 2d\} \gg d^{1/\alpha}$, while $\delta_d = d + \sigma_d/2 \ll d^{1+1/\alpha}$, so the general term is $\ll 2^{-c' d^{1/\alpha}} d^{2+2/\alpha}$, which is summable: an exponential in a positive power of d beats a polynomial. The bound involves α and c and not k. $\square$

Both halves are visible in the computation. For $\alpha = 1/2$ the series reads 4923, 10245, 16608, 24464 at $k = 60, 100, 140, 180$, growing like $k^{3/2}$ — which is $a_1^3 = k^{3(1-\alpha)}$ with $\alpha = 1/2$, exactly as (i) predicts. For $\alpha \ge 1$ it is flat in k to every digit printed:

profile	odds	$a_i \asymp i^{3/2}$	squares	cubes	$\alpha = 1/2$
α	1	3/2	2	3	1/2
$\sum_d 2^{-N_d}\delta_d^2$	69.5	266.07	6473.4	$1.112 \cdot 10^7$	*grows*

[experimentally confirmed, $k = 60, \ldots, 180$; `alpha_r092`] Proposition 7.1 is about clean powers, and the primes do not route through it: Theorem 4.1 for the primes quotes §5.2, and the primes' (H) is checked directly where (H) is stated, from $N_d = \pi(2d) - 1 \gg d/\log d$. No reader need look for $i \log i$ in part (ii). So **(H) cuts the power profiles at exactly** $\alpha = 1$, and the cubes sit inside the admissible range but beyond the reach of computation.

8 The biased invariant

Let P_q be the product measure under which each $a_i \in A$ is present independently with probability $q \in (0,1)$, and write $\mathrm{lm}_q(n)$, $r_q(n)$ for the P_q-measures of the sets of strict local minima and of ground states at the target n. The target is taken at the q-tilted centre $n_q = \lfloor qT \rceil$.

Definition 8.1 (biased gap series)**.**

$$\Gamma^{(q)}(A) \;=\; 1 + \sum_{d\geq 1}\Big[q^{N_d} + (1-q)^{N_d}\Big].$$

At $q = \frac{1}{2}$ every bracket is $2 \cdot 2^{-N_d}$ and $\Gamma^{(1/2)}$ is the window series W_D of Part I [3], hence $\Gamma(A) + (2D+1)2^{-k}$.

Remark 8.2 (why the classification survives the deformation). Part I's classification is measure-free: at error d, an overshooting configuration is a strict local minimum exactly when every $a \leq 2d$ is *excluded*, and an undershooting one exactly when every $a \leq 2d$ is *included*. Under P_q those two events have probability $(1-q)^{N_d}$ and q^{N_d}, which is Definition 8.1. Writing $r^q_B(m) = \sum_{U\subseteq B,\ \sigma(U)=m} q^{|U|}(1-q)^{|B|-|U|}$ and factorising P_q across I_d and B_d,

$$\mathrm{over}_q = (1-q)^{N_d} r^q_{B_d}(n+d), \quad \mathrm{under}_q = q^{N_d} r^q_{B_d}(n-d-\sigma_d), \quad r_q = \sum_{J\subseteq I_d} q^{|J|}(1-q)^{N_d-|J|} r^q_{B_d}(n-\sigma(J)),$$

and the binomial sum over J is exactly 1, so a flat r^q gives the d-th term of Definition 8.1. [derived; the assignment verified per stratum, see Remark 8.8]

8.1 The fair coin is the flattest

Lemma 8.3 (termwise minimality)**.** *For every integer $N \geq 0$ the function $f_N(q) = q^N + (1-q)^N$ on $(0,1)$ is symmetric about $q = \frac{1}{2}$ and convex, and $f_N(q) \geq f_N(\frac{1}{2}) = 2^{1-N}$, with strict inequality for $q \neq \frac{1}{2}$ as soon as $N \geq 2$.*

Proof. $f_N(1-q) = f_N(q)$ is immediate. For $N \geq 2$, $f'_N(q) = N\big[q^{N-1} - (1-q)^{N-1}\big]$ and $f''_N(q) = N(N-1)\big[q^{N-2} + (1-q)^{N-2}\big] > 0$, so f_N is strictly convex; and $t \mapsto t^{N-1}$ is strictly increasing, so $f'_N < 0$ on $(0, \frac{1}{2})$ and $f'_N > 0$ on $(\frac{1}{2}, 1)$. For $N \in \{0, 1\}$, f_N is the constant 2 or 1. ∎

Remark 8.4 (formally verified). Lemma 8.3 is machine-checked. `termwise_min` gives the inequality for every N and every $q \in [0,1]$, `termwise_min_strict` its strictness for $N \geq 2$ and $q \neq \frac{1}{2}$, and `termwise_min_iff` the equality case as a biconditional; the normalisation $2 \cdot (\frac{1}{2})^N = 2^{1-N}$ is `two_mul_half_pow`. The non-strict bound is proved twice, once from convexity of x^N on $[0, \infty)$ and once from Bernoulli's inequality (`termwise_min'`), so the two proofs check the *statement* against each other and not merely the tactic script. No `sorry`, and no axioms beyond `propext`, `Classical.choice`, `Quot.sound`. [Lean-verified; `Pnp/Theory/TermwiseMin.lean`]

Theorem 8.5 (the fair coin minimises the invariant)**.** *For every finite increasing sequence A of odd positive integers with $|A| \geq 2$ and every $q \in (0,1)$,*

$$\Gamma^{(q)}(A) \;\geq\; \Gamma^{(1/2)}(A),$$

with equality if and only if $q = \frac{1}{2}$.

Proof. Sum Lemma 8.3 over d. For strictness it suffices that some $d \leq D$ has $N_d \geq 2$: take any $d \geq (a_2 - 1)/2$, which is at most $D = (a_k - 1)/2$ because $a_2 \leq a_k$. ∎

Remark 8.6 (what the theorem says). $q(1-q)$ is the per-element variance of the coin. Biasing the coin in either direction lowers that variance and raises the invariant: **the more predictable the coin, the more rugged the landscape**. The effect is exactly symmetric in $q \leftrightarrow 1-q$, as it must be, since complementing every configuration exchanges q with $1-q$ and n with $T-n$.

Corollary 8.7 (closed form for the odd numbers)**.** *For $A = (1, 3, \ldots, 2k-1)$ one has $N_d = d$ for $d \le k$, so in the limit $k \to \infty$*

$$\Gamma^{(q)} = 1 + \frac{q}{1-q} + \frac{1-q}{q} = \frac{1}{q(1-q)} - 1 \ \ge\ 3,$$

with equality only at $q = \frac{1}{2}$: the invariant of the odd numbers is the reciprocal of the coin's variance, minus one. *[verified against the summed series to $5.3 \cdot 10^{-15}$ over $q = 0.10, \ldots, 0.80$; e5_r098]*

8.2 Verification

[experimentally confirmed] Measured lm_q/r_q at n_q against the point prediction of Definition 8.1, relative error:

profile	q	$k = 60$	$k = 100$	$k = 140$
odds	0.20	$1.00 \cdot 10^{-2}$	$3.16 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$
odds	0.30	$2.63 \cdot 10^{-3}$	$8.02 \cdot 10^{-4}$	$3.75 \cdot 10^{-4}$
odds	0.50	$2.73 \cdot 10^{-4}$	$5.99 \cdot 10^{-5}$	$2.19 \cdot 10^{-5}$
primes	0.30	$2.18 \cdot 10^{-3}$	$5.02 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$
primes	0.50	$1.73 \cdot 10^{-4}$	$2.26 \cdot 10^{-5}$	$7.32 \cdot 10^{-6}$

There are no free parameters. Two features: the error decays faster than $1/k$ (fitted exponent 2.25 at $q = 0.2$ to 3.0 at $q = 0.5$), and it grows monotonically as q leaves $\frac{1}{2}$, by a factor ≈ 40 from $q = 0.5$ to $q = 0.2$ at fixed k. The second is what the biased flatness hypothesis will have to quantify. [**TODO** state the biased (H); design after §8.3.]

Remark 8.8 (the assignment, and a test that could not decide it)*.* Definition 8.1 is symmetric in $q \leftrightarrow 1-q$, and so is the exact complementation identity $\mathrm{lm}_q(n) = \mathrm{lm}_{1-q}(T-n)$ — at the q-tilted centre, $T - qT = (1-q)T$. Neither can therefore distinguish the assignment of q^{N_d} and $(1-q)^{N_d}$ to the two strata: measured at $k = 100$, the ratio at $q = 0.4$ and at $q = 0.6$ agrees to every printed digit. The discriminating test is per stratum, over_q/r_q against $(1-q)^{N_d}$ and under_q/r_q against q^{N_d}; measured, both ratios lie in $[0.9978, 1.0018]$ over $d \le 6$, whereas a swap would disagree by a factor of 161 at $d = 6$. [experimentally confirmed; e5_r098]

8.3 The bias breaks the reflection symmetry

[experimentally confirmed, shape] Off the q-tilted centre the deviation $\mathrm{dev}_q = (\mathrm{lm}_q/r_q)/\Gamma^{(q)} - 1$ behaves differently at $q = \frac{1}{2}$ and away from it, and the reason is structural. The transfer-function strand's transfer function is even in λ because the two strata sit at $\pm\delta_d$ about a common centre — the analytic form of $\mathrm{lm}(n) = \mathrm{lm}(T-n)$. Under bias that symmetry maps q to $1-q$ rather than to itself, so the odd orders need not cancel, and the first-order coefficient is

$$L^{(q)}(A) \ = \ \sum_{d \ge 1} \delta_d \Big[q^{N_d} - (1-q)^{N_d} \Big],$$

which is antisymmetric in $q \leftrightarrow 1-q$ and vanishes identically at $q = \frac{1}{2}$. Measured for the odd numbers at $k = 80$: $L^{(q)} = -28.634, -8.843, 0, +8.843, +28.634$ at $q = 0.3, 0.4, 0.5, 0.6, 0.7$, and for the primes $-83.717, -23.677, 0, +23.677, +83.717$.

The data agree. Over a range of targets, the spread (max over min) of $|\mathrm{dev}_q - \mathrm{dev}_q(0)|$ divided by λ_q and by λ_q^2:

profile	q	divided by λ_q	divided by λ_q^2
odds	0.30	**1.055**	3.778
odds	0.50	3.118	**1.002**
primes	0.30	**1.162**	4.149
primes	0.50	3.104	**1.012**

So the response is linear in λ under bias and quadratic at $q = \frac{1}{2}$, where $(\mathrm{dev} - \mathrm{dev}_0)/\lambda^2$ is constant to three digits — 19.8221 to 19.8586 for the odd numbers and 48.8625 to 49.4422 for the primes, across a threefold range of the target offset.

Remark 8.9 (the range is part of the test). The first measurement used offsets $x \in [0.04, 0.20]$ and came out incoherent for the primes at $q = \frac{1}{2}$ (a quadratic spread of 15.4). The reason is worth recording because it is a property of ratio tests and not of this problem. At $q = \frac{1}{2}$ the response is *even* in x, so $\mathrm{dev} - \mathrm{dev}_0$ cannot change sign for the reason under test; at finite k the vertex of the parabola sits at some $x_* \neq 0$, and $\mathrm{dev} - \mathrm{dev}_0$ crosses zero at $\pm x_*$. Measured, $x_* \approx 0.03$, with the signs symmetric in $\pm x$ exactly as that account predicts. A range containing the crossing makes *both* ratios pass through zero, so the spread of either is set by how close a grid point landed to x_* — an accident of the grid. **A ratio test must not straddle a zero of its own numerator.** The table above uses $x \in [0.10, 0.30]$, outside x_*; the repair is the range, not the omission of the offending point.

[experimentally confirmed; e6_r100, e6b_r115, whose pipeline is checked against e6_r100 by reproducing $[19.8186, 19.8352]$ on the original range, and whose discriminating control at $q = 0.3$ still returns *linear* on the new range]

9 The biased transfer function

9.1 The sign of λ, which now matters

The transfer-function strand's transfer function is even in λ, so the sign of the slope is not observable there. Splitting the cosh makes it observable, so we fix it first:

$$\lambda_q(n) \;=\; +\frac{d}{dm} \log r_A^q(m)\Big|_{m=n} \;=\; \tfrac{1}{4} \log \frac{r_A^q(n+2)}{r_A^q(n-2)} + O(\cdot),$$

positive below the biased centre. This is the convention under which $\lambda \approx s$; the transfer-function strand's 2.1 carried the opposite sign and has been corrected, with no numerical consequence there for exactly the reason just given.

9.2 The derivation

Take $P_{q,s}$: element a present with weight $q\, e^{-sa}$, absent with weight $1-q$, so $p_a = qe^{-sa}/((1-q) + qe^{-sa})$ and $p_a = q - q(1-q)sa + O(s^2a^2)$. Two consequences, both absent at $q = \frac{1}{2}$:

- the per-element variance is $a^2 p_a(1-p_a) = q(1-q)a^2(1 + O(sa))$, so the layer weight of the transfer-function strand's 5.21 becomes $e^{-\lambda^2 q(1-q)s_d/2}$;
- removing I_d removes mean $q\sigma_d$, so the two strata no longer sit at $\pm\delta_d$ about a common centre. They sit at

$$\delta_d^+ = d + q\sigma_d, \qquad -\delta_d^-, \quad \delta_d^- = d + (1-q)\sigma_d.$$

The bias splits the pair asymmetrically, in the ratio $q : (1-q)$.

Hence

$$\Phi^{(q)}(\lambda) \;=\; 1+\sum_{d\geq 1}\Big[(1-q)^{N_d}e^{\lambda\delta_d^+}+q^{N_d}e^{-\lambda\delta_d^-}\Big]e^{-\lambda^2 q(1-q)s_d/2}. \tag{4}$$

[derived; three internal checks below are exact, the point prediction is verified]

Three checks are built in, and all three hold to machine zero ($0.00{\cdot}10^0$, two profiles, $k=60$): $\Phi^{(q)}(0)=\Gamma^{(q)}$ of Definition 8.1; the complementation symmetry $\Phi^{(q)}(\lambda)=\Phi^{(1-q)}(-\lambda)$, which holds because $\delta^+(q)=\delta^-(1-q)$; and $\Phi^{(1/2)}$ is the transfer-function strand's Φ term by term. [experimentally confirmed; `e10_r105`]

9.3 Tilted rigidity

Proposition 9.1 (the biased landscape responds at first order)**.** $\Phi^{(q)\prime}(0)=\sum_{d\geq 1}\Big[(1-q)^{N_d}\delta_d^+ - q^{N_d}\delta_d^-\Big]$, *which vanishes identically at $q=\frac{1}{2}$ and is non-zero otherwise. So rigidity at a biased measure is not flatness: it is a* tilt *of a definite slope, and the slope is a point prediction with no free parameter.*

[derived and verified, not proved from a hypothesis-free statement: it is a consequence of $\Phi^{(q)}$, itself derived by the partition-function argument of §9 and confirmed by three exact internal checks. The first-order coefficient is a point prediction with no free parameter, and §9.4 tests it against the naive alternative on 36 discriminating rows — which is why the two are distinguished here rather than only in §14.]

This is what §8.3 demanded of the derivation, and it is a sharper demand than it looks. The naive guess — keep the transfer-function strand's $\delta_d=d+\sigma_d/2$ and merely split the weights — gives instead $\sum[(1-q)^{N_d}-q^{N_d}]\delta_d$, and the two differ by exactly $(\frac{1}{2}-q)\sum_d\sigma_d[q^{N_d}+(1-q)^{N_d}]$: zero at $q=\frac{1}{2}$, and 20–32% of the prediction at the q tested. The measurement therefore decides between them.

9.4 Verification

The estimator must kill constants, because the point prediction of §8.2 carries an $O(10^{-3})$ offset at the biased centre which is the same size as the linear term. So we use a central difference about n_q: with $n_\pm=\lfloor(q\pm hq)T\rceil$,

$$\widehat{L}(h)=\frac{(\mathrm{lm}_q/r_q)(n_+)-(\mathrm{lm}_q/r_q)(n_-)}{\lambda_q(n_+)-\lambda_q(n_-)}.$$

Ratios of $\widehat{L}$ to the two candidates, at $h=0.01$:

profile	q	separation	$\widehat{L}/\Phi^{(q)\prime}(0)$	$\widehat{L}$/naive
odd numbers, $k=120$	0.30	31.6%	**0.9945**	0.6805
odd numbers, $k=120$	0.40	19.9%	**0.9976**	0.7991
odd numbers, $k=120$	0.50	—	$\Delta(\mathrm{lm}_q/r_q)=0$ exactly	
odd numbers, $k=120$	0.60	19.9%	**0.9976**	0.7991
primes, $k=120$	0.30	32.3%	**0.9981**	0.6754
primes, $k=120$	0.40	20.2%	**0.9983**	0.7967

Over all 36 discriminating rows (two profiles, $k=80,120$, four q, four h), the median $|\widehat{L}/\Phi^{(q)\prime}(0)-1|$ is 0.0041 against 0.2038 for the naive candidate. The one row above 10% is the primes at $k=80$, $q=0.3$, $h=0.01$, where $\Delta\lambda=8.5\cdot 10^{-5}$ and the integer rounding of $n_\pm$ is a visible fraction of the step; the same row at $k=120$ reads 0.9981. At $q=\frac{1}{2}$ the numerator is 0 or $4.4 \cdot 10^{-16}$, below the stated floor — the fair coin has no linear response, which is the content of Proposition 9.1 at its zero. [experimentally confirmed; e10_r105]

The split is also visible stratum by stratum, which tests the offsets directly rather than through their sum: at $k = 80$, $q = 0.3$, over_q/r_q against $(1-q)^{N_d} e^{\lambda\delta_d^+} e^{-\lambda^2 q(1-q)s_d/2}$ agrees to within 0.24% over $d = 1, \dots, 6$, and under_q/r_q against its partner to within 0.9%. [experimentally confirmed; e10_r105]

9.5 The biased hypothesis

(H_q): $\sum_d [(1-q)^{N_d}(\delta_d^+)^2 + q^{N_d}(\delta_d^-)^2] < \infty$, uniformly for $q \in [q_0, 1-q_0]$. Since $\max(q, 1-q) \le 1 - q_0 < 1$, the geometric damping of the transfer-function strand's (H) survives with base $1 - q_0$, and the constant degrades as $q_0 \to 0$ — which is the $\sim 40\times$ growth of the §8.2 error already measured between $q = \frac{1}{2}$ and $q = 0.2$. The $\alpha = 1$ threshold of the transfer-function strand is unchanged, the damping argument being base-agnostic. [derived; the degradation matches the measured trend]

10 Random sequences, unconditionally

Before the theorem, the floor it has to stand on. Let A be drawn from the Cramér model on the odd integers: each odd $m \in [3, N]$ is included independently with probability $2/\log m$, and N is fixed by $\mathbb{E}|A| = k$. Nothing is conditioned, so $|A|$ fluctuates from instance to instance. That is deliberate: $kR_A(k)$ is the k-invariant quantity of the transfer-function strand's 5.7, so an ensemble of genuinely different sizes tests the $1/k$ law for free. Twenty instances at each of two sizes, each through the whole paper-3 pipeline with its own N_d and σ_d:

$\mathbb{E}\lvert A\rvert$	range of k	(H) series	mean kR	sd	sd/mean
120	101–135	133.79–320.79	214.96	10.58	4.92%
180	146–191	132.94–471.77	206.32	6.94	3.37%

at $x = 0.06$; $c_A = kR/100$. Three things, and the third is the one to keep. [experimentally confirmed; e7_r102]

(i) The hypothesis holds on the ensemble. All forty instances have a bounded (H) series and a window that tracks the primes'. This is a precondition, not a result: a theorem about random sequences is unconditional only if the random sequences satisfy (H) themselves.

(ii) The constant concentrates. The spread across instances is under 5% and falls with k, while the mean of kR moves by 4% between the two sizes — the $1/k$ law of the transfer-function strand surviving an ensemble it was not fitted on.

(iii) The primes sit inside the ensemble and the odd numbers do not. Against the $k \approx 180$ ensemble (2.0632 ± 0.0694 in c_A), the primes score $z = -0.39$ and the odd numbers $z = -2.70$, at every one of the four targets. In this statistic the primes are indistinguishable from a random odd sequence of the same density; the odd numbers are a different object.

There is one place where the random model is *not* like the primes, and it is the place that matters for the minor arcs. Write $h_d(q) = \max_{(r,q)=1} \left(\prod_{a \in B_d} |\cos(\pi r a/q)|\right)^{1/|B_d|}$. Since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a, the modulus-4 floor $h_d(4) = 1/\sqrt{2}$ is deterministic, not merely universal. Part II [4]'s vanishing lemma kills the modulus-$2m$ peak (m odd) as soon as B_d contains an element $\equiv 3 \pmod 6$. A set of primes contains exactly one such element, namely 3, and 3 leaves B_d at $d \ge 2$, so the primes keep $h_d(6) = \sqrt{3}/2$; a random odd sequence contains about $|B_d|/3$ of them and loses the peak at every d. Measured over

$q \le 120$ at $d = 1, 2, 5$: every random instance has $\max_{q\ge3} h_d(q) = 1/\sqrt{2}$ attained at $q = 4$, and the primes have $\sqrt{3}/2$ at $q = 6$. [experimentally confirmed; e7_r102]
So the random theorem should come out with the geometric rate $1/\sqrt{2} = 0.7071$ per element rather than the $\sqrt{3}/2 = 0.8660$ of Parts I and II [3, 4]: *the primes are the harder case, and the modulus-6 ripple is a structural feature of the primes that the random model destroys.*

10.1 The modulus-4 theorem

The observation above is not an accident of the ensemble; it is a theorem about residues, and it is the exact counterpart of Part II's modulus-6 theorem for a different equidistribution class. Part II averages $|\cos|$ over the *reduced* residues, which is the right class for the primes. A random odd sequence is equidistributed over the *odd* residues instead, so the relevant quantity is a different one.

Definition 10.1. For $q \ge 2$ let $R_q = \mathbb{Z}/q$ if q is odd and $R_q = \{j : j \text{ odd}\} \subset \mathbb{Z}/q$ if q is even — for q even and m odd, $m \bmod q$ is odd, so R_q is the set of residues an odd sequence can occupy. Put

$$M_{\mathrm{odd}}(q) \;=\; \Big(\prod_{j\in R_q} |\cos(\pi j/q)|\Big)^{1/|R_q|}.$$

For r coprime to q the map $j \mapsto jr$ permutes R_q, so $M_{\mathrm{odd}}(q)$ does not depend on the numerator of the rational r/q.

Everything below is a corollary of the coset identity of §3, because $\{j/q : j \in R_q\}$ *is a* coset of $(1/v)\mathbb{Z}$.

Corollary 10.2 (the identity in the form §10.2 uses). *Let $q \ge 2$, put $v = q$ for q odd and $v = q/2$ for q even, and let $\sigma = \frac{1}{2}$ if $q \equiv 2 \pmod 4$ and $\sigma = 0$ otherwise. Then $|R_q| = v$ and, for every real s,*

$$\prod_{j\in R_q} \Big|\cos\pi\Big(s+\frac{j}{q}\Big)\Big| \;=\; 2^{\,1-v}\,|\cos\pi(vs+\sigma)|, \qquad \textit{hence} \qquad \frac{1}{v}\sum_{j\in R_q} X\Big(s+\frac{j}{q}\Big) \;\ge\; \Big(1-\frac{1}{v}\Big)\log 2.$$

Proof. $\{j/q : j \in R_q\}$ is $\{k/v\}_{k<v}$ for q odd and $\{\frac{1}{2v} + \frac{k}{v}\}_{k<v}$ for q even. Apply Lemma 3.1 at $t = s$ and at $t = s + \frac{1}{2v}$ respectively. In the second case the shift contributes $v \cdot \frac{1}{2v} = \frac{1}{2}$ to the argument, so the total offset is $\tau_v + \frac{1}{2}$, which is an integer when v is even and $\frac{1}{2}$ when v is odd; and v odd with q even is exactly $q \equiv 2 \pmod 4$. The inequality is Corollary 3.2. ☐

Lemma 10.3 (closed form). *$M_{\mathrm{odd}}(q) = 2^{1/q-1}$ for q odd. For $q = 2u$ even, $M_{\mathrm{odd}}(q) = 2^{1/u-1}$ if u is even, and $M_{\mathrm{odd}}(q) = 0$ if u is odd.*

Proof. Put $s = 0$ in Corollary 10.2: the right-hand side is 2^{1-v} when $\sigma = 0$ and 0 when $\sigma = \frac{1}{2}$. Take v-th roots. ☐

Remark 10.4 (a second proof of the odd case, with no roots of unity). For q odd the closed form also follows from the double-angle formula alone, and this is the route we formalise. Let $P = \prod_{j=1}^{q-1} |\sin(\pi j/q)|$ and $C = \prod_{j=1}^{q-1} |\cos(\pi j/q)|$. From $\sin 2x = 2\sin x\cos x$, $\prod_{j=1}^{q-1} |\sin(2\pi j/q)| = 2^{q-1}PC$. Since q is odd, $j \mapsto 2j \bmod q$ is a bijection of $\{1, \ldots, q-1\}$ and $|\sin(2\pi j/q)| = |\sin(\pi\,(2j \bmod q)/q)|$, so the left side is P again. As $P \ne 0$, $C = 2^{1-q}$, and $\prod_{j\in R_q} |\cos(\pi j/q)| = C$ because the $j = 0$ factor is 1. Two proofs of one load-bearing identity check the *statement*; one proof checks only the argument. This is the branch that is machine-checked, as `prod_abs_cos_odd`; see Remark 10.6.

Theorem 10.5 (the modulus-4 theorem)**.** $\max_{q \geq 2} M_{\mathrm{odd}}(q) = \dfrac{1}{\sqrt{2}}$*, attained at* $q = 4$ *and nowhere else.*

Proof. $v \mapsto 2^{1/v-1}$ is strictly decreasing. For q odd, $q \geq 3$, Lemma 10.3 gives $M_{\mathrm{odd}}(q) = 2^{1/q-1} \leq 2^{-2/3}$. For $q \equiv 2 \pmod 4$ it gives 0. For $q \equiv 0 \pmod 4$, write $q = 2u$ with u even, $u \geq 2$; then $M_{\mathrm{odd}}(q) = 2^{1/u-1} \leq 2^{-1/2}$ with equality exactly when $u = 2$, i.e. $q = 4$. And $2^{-2/3} < 2^{-1/2}$. □

Lemma 10.3 is verified against the product for all $q \leq 200$, worst discrepancy $4.4 \cdot 10^{-16}$; Corollary 10.2 is verified in its product form for $q \leq 120$ at 400 random shifts each, worst discrepancy $4.4 \cdot 10^{-16}$, and independently at 60 decimal digits, worst discrepancy $2.2 \cdot 10^{-61}$. [proved; and checked numerically, `mq4_r107`, `stab_r110`]

Remark 10.6 (what is machine-checked here, and what is not)**.** The two halves of the argument have different statuses and the difference is worth stating. *The maximisation is formally verified.* Taking the closed form of Lemma 10.3 as a definition, `ModdCF_le` gives $M_{\mathrm{odd}}(q) \leq 2^{-1/2}$ for every $q \geq 2$, `ModdCF_eq_iff` the equality case as a biconditional, `ModdCF_lt_of_ne_four` the strict bound off $q = 4$ by a second route, and `two_rpow_neg_half` the normalisation $2^{-1/2} = 1/\sqrt{2}$. This is the step where an error would hide — a case split over three residue classes and the comparison of $2^{-2/3}$ with $2^{-1/2}$ — and it is now checked by machine. *Of the closed form itself, the odd branch is now formalised and the even branch is not.* `prod_abs_cos_odd` proves $\prod_{j=1}^{q-1} |\cos(\pi j/q)| = (1/2)^{q-1}$ for every odd q, by the double-angle route of Remark 10.4 — which is exactly why that route is worth having: it uses one bijection and one trigonometric identity, where the proof of Lemma 3.1 needs a product over roots of unity that Mathlib does not carry. The even branch, and Lemma 3.1 in general, remain proved on paper and checked numerically. [Lean-verified (maximisation, and the odd closed form); `Pnp/Theory/ModFour.lean`, `Pnp/Theory/OddProd.lean`]

Remark 10.7 (the two classes, side by side).

	class of residues	extremal q	rate per element
primes (Part II)	reduced	6	$\sqrt{3}/2 = 0.8660$
random odd A (here)	odd	4	$1/\sqrt{2} = 0.7071$

The two disagree exactly at $q \equiv 2 \pmod 4$: $M(6) = \sqrt{3}/2$ but $M_{\mathrm{odd}}(6) = 0$, because $j = 3$ is an odd residue mod 6 and $\cos(\pi/2) = 0$. A set of primes contains at most one element $\equiv 3 \pmod 6$, and it leaves B_d at $d \geq 2$; a random odd sequence contains a positive proportion of them at every d. *So the primes are the harder case, by a factor* $\sqrt{3/2} = 1.2247$ *in the geometric rate*, and Part II's $\sqrt{3}/2$ is a feature of the primes rather than a limitation of the method.

Both halves of that comparison are machine-checked. `abs_cos_pi_mul_div_four` gives $|\cos(\pi a/4)| = 1/\sqrt{2}$ for *every* odd a, so `prod_abs_cos_four` makes the modulus-4 floor $(1/\sqrt{2})^{|A|}$ *deterministic* rather than merely universal; and `cos_pi_mul_div_six_eq_zero` gives $\cos(\pi a/6) = 0$ whenever $a \equiv 3 \pmod 6$, so `prod_abs_cos_six_eq_zero` kills the modulus-6 peak on a single element. `six_dies_four_survives` is the separation itself: for an odd sequence containing such an element the modulus-6 product is 0 while the modulus-4 product is still exactly $(1/\sqrt{2})^{|A|} > 0$. [Lean-verified; `Pnp/Theory/OddPeaks.lean`]

10.2 The rate, and what is still missing

Theorem 10.8 (minor-arc rate for random odd sequences)**.** *Let A be drawn from the Cramér model above and let $F_A(\theta) = \prod_{a \in A} |\cos(\pi a\theta)|$. Then almost surely*

$$\lim_{b \to \infty} \max_{\theta \text{ minor}} F_A(\theta)^{1/b} = \frac{1}{\sqrt{2}}.$$

Here minor *means* $\|\theta\| \geq 1/N$, *with* $\|\cdot\|$ *the distance to the nearest integer; Step* $2''$ *shows that this radius is large enough, so nothing is left unowned.*

[proved, with every constant written out. The proof is the four steps below rather than a `proof` environment, so it is worth saying here that it is one: no step is deferred, no constant is fitted, and the only external input is the arithmetic half, Theorem 10.5, which is proved in this paper. See Remark 10.10 for what an independent reader should check first.]

The arithmetic half is Theorem 10.5. Here is the rest, in the steps it takes, with the constants named. Fix the Dirichlet parameter $Q = \lfloor\sqrt{N}\rfloor$ — Step $2'$ says why that value and not another — and set

$$M = \log\frac{2Q}{\pi} + 1, \qquad \boxed{\delta = \tfrac{1}{6}\log 2 = 0.1155245\ldots}$$

Write $X_m(\theta) = -\log|\cos(\pi m\theta)| \in [0,\infty]$, so that $-\log F_A(\theta) = \sum_m \xi_m X_m(\theta)$ with ξ_m the independent indicators of the model. Only a *lower* bound on this sum is wanted, which is what makes the argument easy.

Step 1 (truncate; the unbounded tail is free). Put $X_m^{(M)} = \min(X_m, M)$. Then $\sum \xi_m X_m \geq \sum \xi_m X_m^{(M)}$, and the summands now lie in $[0, M]$. At $\theta = \frac{1}{4}$ truncation is inactive, since $X_m(\frac{1}{4}) = \frac{1}{2}\log 2$ for every odd m.

Step 2 (the mean, and the constant δ). At $\theta = r/q$ the values X_m over odd m run through $\{-\log|\cos(\pi j/q)| : j \in R_q\}$, so by Lemma 10.3 their mean is $-\log M_{\mathrm{odd}}(q) = (1 - 1/v)\log 2$ and the *gap* over the value $\frac{1}{2}\log 2$ at $q = 4$ is

$$\delta(q) \;=\; \left(\tfrac{1}{2} - \tfrac{1}{v}\right)\log 2, \qquad v = q\ (q \text{ odd}), \quad v = q/2\ (q \equiv 0 \bmod 4), \quad \delta = \infty\ (q \equiv 2 \bmod 4).$$

This vanishes exactly at $v = 2$, i.e. $q = 4$, and increases with v; the smallest admissible v after 2 is $v = 3$. **So the binding competitor is $q = 3$ and the gap is $\delta = \frac{1}{6}\log 2$**, with $q = 8$ next at $\frac{1}{4}\log 2$. Step $2'$ will replace the entry $\delta = \infty$ at $q \equiv 2 \pmod 4$ by the finite value $(\frac{1}{2} - \frac{1}{u})\log 2$: off the rational point the vanishing residue no longer vanishes, and $q = 6$ then *ties* with $q = 3$ at $\frac{1}{6}\log 2$. The constant is unchanged and two moduli bind it rather than one. The two moduli with $v \leq 2$ — $q = 4$, where the floor is $\frac{1}{2}\log 2$ exactly and is meant to be, and $q = 2$, where $v = 1$ and the floor degenerates to 0 — are handled in Step $2''$, together with $q = 1$. [proved from Lemma 10.3; checked against the direct residue average for all $q \leq 400$, worst discrepancy $1.0 \cdot 10^{-15}$, and the minimum located at $q = 3$ over $q \leq 2000$; `delta_r109`]

Truncation does not eat this. For $q \not\equiv 2 \pmod 4$ one has $|\cos(\pi j/q)| \geq \sin(\pi/2q)$, so $\max_j X_m \leq \log(2q/\pi) < M$ and truncation is *inactive* at every $q \leq Q$ — measured, the largest value over all such $q \leq 1000$ is 6.4552 against $M - 1 = 6.4562$, so the bound $\log(2Q/\pi)$ is sharp and the $+1$ in M is there to keep the argument off a knife edge. For $q = 2u \equiv 2 \pmod 4$ truncation *is* active, at the single odd residue $j = u$ where the cosine vanishes; putting $x = -1$ in $(x^u + 1)/(x + 1)$ gives $\prod_{j \neq u}|\cos(\pi j/q)| = u\,2^{1-u}$, so the truncated mean is $[(u-1)\log 2 - \log u + M]/u$, which exceeds $\frac{1}{2}\log 2 + \delta$ by between 0.23 and 1.25 over $q \leq 198$ at $Q = 200$. [experimentally confirmed, exact to $4.4 \cdot 10^{-16}$; `delta_r109`]

Step $2'$ (from the rationals to every θ: there is no dent). Write $\theta = a/q + \beta$ with $q \leq Q$ and $|\beta| \leq 1/(q(Q+1))$ (Dirichlet). The obstacle here looks like a job for a quantitative equidistribution estimate — Erdős–Turán or Koksma — and it is not: Lemma 3.1 settles it as an *identity*. Since $j \mapsto ja$ permutes R_q, the values $\{ja/q\}$ are the $\{j/q\}$, and for any common shift s

$$\frac{1}{|R_q|}\sum_{j\in R_q} X\Big(s + \frac{j}{q}\Big) \;=\; \underbrace{\Big(1 - \frac{1}{v}\Big)\log 2}_{-\log M_{\mathrm{odd}}(q)} + \frac{1}{v}X(vs + \sigma) \;\geq\; -\log M_{\mathrm{odd}}(q), \qquad (*)$$

because $X \geq 0$. *Shifting the whole rational grid can only raise the mean*; the dent one expects when θ moves off a rational does not exist, and no discrepancy theory is needed to bound it. Equality in $(*)$ holds exactly on $vs + \sigma \in \mathbb{Z}$, so the rational values are the true minima, not merely convenient sample points. [proved (Lemma 3.1); the minimum over s of the left side of $(*)$ is checked to equal $-\log M_{\text{odd}}(q)$ to $3 \cdot 10^{-16}$ for 15 moduli, and to survive truncation at M for every $q \leq 1000$; `stab_r110`]

First, truncation does not spoil $(*)$. The loss $(X - M)^+$ is nonzero only within e^{-M}/π of a pole, and the coset points are $1/v \geq 1/Q > 1/(Qe) = 2e^{-M}/\pi$ apart, so at most one residue is affected; if it lies at distance d from the pole then $X \leq \log \frac{1}{2d}$, while the excess term of $(*)$ is $\frac{1}{v} X(vs + \sigma) \geq \frac{1}{v} \log \frac{1}{\pi v d}$ because $vs + \sigma$ sits at distance vd from a pole. The floor therefore survives as soon as $M \geq \log(\pi v/2)$, and with $M = \log(2Q/\pi) + 1$ and $v \leq Q$ that reads $1 \geq \log(\pi^2/4)$ — true, **with a margin of exactly** $1 - \log(\pi^2/4) = 0.0968$. This is the third place the $+1$ in M earns itself. [proved; and $\min_s \Phi_q^{(M)}(s) - (1 - 1/v) \log 2$ is checked to be $\geq -5.6 \cdot 10^{-16}$ for every $q \leq 1000$; `stab_r110`, `rate_r112`]

Step $2'$ continued (the two error terms, in full). What $(*)$ does not by itself cover is that element m of the class j sits at $ja/q + m\beta$, so the shift varies *within* a class. Fix $q \notin \{1, 2\}$ and group the odd $m \in [3, N]$ by residue class mod q. In a class the m form an arithmetic progression of common difference $2v$ (for q odd the odd numbers in a class mod q step by $2q = 2v$; for $q = 2u$ they step by $q = 2v$), so the phases form an arithmetic progression of step $\eta = 2v\beta$ sweeping an arc of length $L = N|\beta|$, and the class starts at $x_j + m_j \beta$ where $m_j \leq 2v$ is its first element. Three errors, and nothing else:

(i) Progression versus integral. $X^{(M)}$ is of bounded variation, at most $2M$ per period, so for a progression of step η over an arc of length L, $|\sum_r X^{(M)} - \eta^{-1} \int| \leq 2M(L + 1)$. There are v classes and $\asymp N/2$ terms in all, so in the mean this costs at most $4qM/N + 4M/(Q+1)$, using $v|\beta| \leq 1/(Q + 1)$.

(ii) Jitter. Replacing the arc of class j by the common arc is a translation by $\epsilon = m_j \beta$, and $m_j \leq 2v$ while the progression step is $\eta = 2v\beta$, so $|\epsilon| \leq |\eta|$ *exactly.* Translating by ϵ changes the integral by at most ϵV, so the per-class error is at most $\epsilon V/\eta \leq V \leq 2M(L + 1)$ — the same shape as (i), and not a separate mechanism at all. Per element, $4qM/N + 4M/(Q + 1)$.

(iii) Unequal class counts. They differ by at most 1, costing vM in the sum, i.e. $2qM/N$ in the mean.

After (i)–(iii) the integrand is the honest coset average, which is $\geq (1 - 1/v) \log 2$ pointwise by $(*)$; integrating a pointwise inequality needs no further hypothesis, and in particular *no case distinction on the length of the sweep* — which is what makes the route robust in Q where an arc-comparison design is not. Collecting,

$$\frac{\sum_m p_m X_m^{(M)}(\theta)}{\sum_m p_m} \;\geq\; \left(1 - \frac{1}{v}\right) \log 2 \;-\; E, \qquad E \;\leq\; \frac{10\, qM}{N} + \frac{8M}{Q + 1}.$$

Both terms are minimised together at $Q = \lfloor \sqrt{N} \rfloor$, where $E = O(M/\sqrt{N})$ — which is why the Dirichlet parameter is $\sqrt{N}$ and not, as one first writes, N. At $Q = N$ the argument fails twice over: the residue classes hold $O(1)$ elements each, and $|\beta| \leq 1/(qN)$ makes every sweep shorter than a single period.

The weights need no argument at all. $p_m = 2/\log m$ is *decreasing*, so by Abel summation a bound valid for every prefix $\{m \leq N'\}$ passes to the p-weighted mean unchanged: if $S(N') \geq c\, P(N') - \mathcal{E}(N')$ for all N', then $\sum p_m X_m^{(M)} \geq c \sum p_m - \sum_k (p_k - p_{k+1}) \mathcal{E}(k) - p_N \mathcal{E}(N)$, and with $\mathcal{E}(N') \leq CM(\sqrt{N} + N'/\sqrt{N})$ the subtracted term is $O(M\sqrt{N}) + O(M/\sqrt{N}) \sum p_m$, i.e. $O(M \log N/\sqrt{N})$ relative. There is no block-weight condition to check. [proved; the shortfall at the binding modulus $q = 3$ is measured as $1.9 \cdot 10^{-3}, 5.8 \cdot 10^{-4}, 1.7 \cdot 10^{-4}, 4.8 \cdot 10^{-5}$ at $N = 2001, 8001, 32001, 128001$, and deficit $\cdot\, N/(qM)$ is $0.3057, 0.3140, 0.3199$ at $N =$

$8001, 32001, 128001$ — constant, i.e. the measured deficit has exactly the derived shape qM/N. Errors (i), (ii), (iii) are additionally isolated and measured *one at a time* against their own bounds, over three sizes, eight moduli and three offsets, worst values $1.1 \cdot 10^{-2}$, $1.2 \cdot 10^{-2}$ and $8.5 \cdot 10^{-4}$ against bounds of order 10^{-1}; `stab_r110`, `rate_r112`, `audit_r114`]

Step 2″ (the three moduli with $v \le 2$, and the major arc). Corollary 3.2 gives $\frac{1}{2}\log 2$ at $q = 4$ ($v = 2$) and nothing at all at $q \in \{1, 2\}$ ($v = 1$). The last two are not prose; they are the following lemma, and $q = 4$ needs no more than the identity because $\frac{1}{2}\log 2$ is exactly the value the theorem is about.

Lemma 10.9 (the degenerate moduli)**.** *Let $\phi(w) = -\log|\sin \pi w|$ and $L > 0$. Then $\frac{1}{L}\int_0^L \phi = \log 2 + \mathrm{Cl}_2(2\pi\{L\})/(2\pi L)$, and*

$$\frac{1}{L}\int_0^L \phi \; > \; \tfrac{1}{2}\log 2 \quad \textit{for every } L > 0, \qquad \frac{1}{L}\int_0^L X \; \ge \; \log 2 - \frac{\mathrm{Cl}_2(\pi/3)}{2\pi} = 0.5316 \quad \textit{for } L \ge 1.$$

Consequently the mean over a $q = 2$ sweep exceeds $\frac{1}{2}\log 2$, and so does the mean over a $q = 1$ sweep once $L = N\|\theta\| \ge 1$ — which is exactly the definition of minor.

Proof. The closed form is $\int_0^x \phi = x\log 2 + \mathrm{Cl}_2(2\pi x)/(2\pi)$ together with the periodicity of ϕ. For $L \le \frac{1}{2}$: ϕ is convex on $(0, 1)$, so the average over $[0, L]$ is at least the midpoint value $-\log\sin(\pi L/2) \ge -\log\sin(\pi/4) = \frac{1}{2}\log 2$; at the endpoint $L = \frac{1}{2}$ the two-point midpoint bound $\frac{1}{2}[\phi(L/4) + \phi(3L/4)] = 0.51986$ shows the inequality is strict there. For $L \ge \frac{1}{2}$: $1/(2\pi L) \le 1/\pi$ and $|\mathrm{Cl}_2| \le \mathrm{Cl}_2(\pi/3)$, so the average is at least $\log 2 - \mathrm{Cl}_2(\pi/3)/\pi = 0.3701 > \frac{1}{2}\log 2$. For $q = 1$ the same computation with X in place of ϕ gives $\log 2 + \mathrm{Cl}_2(2\pi L + \pi)/(2\pi L)$, and $L \ge 1$ makes the correction at most $\mathrm{Cl}_2(\pi/3)/(2\pi) = 0.1615$. ☐

The true $q = 2$ floor is 0.4945 at $L = 0.7908$ — above $\frac{1}{2}\log 2 + \delta$, not merely above $\frac{1}{2}\log 2$ — but that is measured, not proved, and the theorem does not need it. The smallest L at which the $q = 1$ average reaches $\frac{1}{2}\log 2 + \delta$ is 0.4528, so the radius $1/N$ in the definition of *minor* is comfortable rather than tight. [proved; the Clausen closed form checked against direct quadrature to $4.5 \cdot 10^{-5}$, and the convexity bound verified at 5000 values of L; `rate_r112`]

Step 3 (concentration). The summands are independent and lie in $[0, M]$, and there are $n = \lfloor (N-1)/2 \rfloor$ of them, so Hoeffding gives $\Pr[\sum \xi_m X_m^{(M)} < \mathbb{E} - t] \le \exp(-2t^2/(nM^2))$. Take $t = \varepsilon\,\mathbb{E}|A|$ and use $\mathbb{E}|A| \asymp 2N/\log N$, $n \asymp N/2$, $Q = \lfloor\sqrt{N}\rfloor$ and hence $M \asymp \frac{1}{2}\log N$: the failure probability at one θ is

$$\exp(-c\,\varepsilon^2 N/\log^4 N), \qquad c = 64.$$

Two readings of the same bound, and the theorem uses the second. With $\varepsilon = \frac{1}{2}\delta$ this is the fixed-margin statement, $\exp(-cN/\log^4 N)$ with $c = 16\delta^2 = 0.2135$. But the theorem only needs the mean to survive to within $o(1)$, so one may instead take $\varepsilon = N^{-1/4}$, which still leaves the exponent $\asymp \sqrt{N}/\log^4 N \to \infty$. *That is why $q \in \{1, 2\}$ need only clear $\frac{1}{2}\log 2$ and not $\frac{1}{2}\log 2 + \delta$*, and it is what makes Lemma 10.9 sufficient. No second moment of an exponential sum appears: the randomness is in the *sequence*, not in the phases.

Step 4 (net, then Borel–Cantelli). Where $X_m < M$ we have $|\cos(\pi m\theta)| \ge e^{-M}$, hence $|\partial_\theta X_m^{(M)}| = \pi m|\tan(\pi m\theta)| \le \pi N e^M \le 2eNQ \le 2eN^{3/2}$, so a net of spacing $\varrho/(2eN^{3/2}b)$ suffices and has $O(N^{5/2}/\varrho)$ points. Since $N^{5/2}\exp(-c\varepsilon^2 N/\log^4 N)$ is summable in N for $\varepsilon = N^{-1/4}$, the union bound and Borel–Cantelli give the almost-sure statement.

Assembling. Steps 1–4 give, almost surely and uniformly over minor θ, $b^{-1}\sum_{a \in A} X_a(\theta) \ge \frac{1}{2}\log 2 - o(1)$, i.e. $\max_{\theta \text{ minor}} F_A(\theta)^{1/b} \le 2^{-1/2}(1 + o(1))$: at $q = 4$ the floor is $\frac{1}{2}\log 2$ by Corollary 3.2, at every other $q \ge 3$ it is $\frac{1}{2}\log 2 + \delta$, at $q \in \{1, 2\}$ it is above $\frac{1}{2}\log 2$ by Lemma 10.9, and the error of Step 2′ is $O(M\log N/\sqrt{N})$. The matching lower bound is free: $\theta = \frac{1}{4}$ is minor and $X_a(\frac{1}{4}) = \frac{1}{2}\log 2$ for *every* odd a, so $F_A(\frac{1}{4})^{1/b} = 2^{-1/2}$ identically. Hence the limit is $2^{-1/2}$. ☐[proved]

Remark 10.10 (what "proved" means here, and what it does not). Theorem 10.8 carries no hypothesis and every constant in its proof is written down. One qualification, stated here rather than left for a referee to raise. The three error terms of Step $2'$ are each reduced to a standard estimate — a Riemann-sum comparison for a function of bounded variation, and Abel summation against a monotone weight — which we state and apply in the form we need but do not prove; a referee would be within rights to ask for them in full, and doing so would lengthen Step $2'$ without changing any constant. *No hypothesis is not the same as written out in full*, a distinction Part II now also makes explicitly about its own main theorem, and the two should be read the same way.

Measured, at $\theta = r/q$ for $q \leq 12$ and eight instances at $b \approx 190$: the mean of $b^{-1} \sum_a -\log|\cos(\pi a/q)|$ agrees with $-\log M_{\text{odd}}(q)$ within the $O(b^{-1/2})$ fluctuation at every q, and at $q = 4$ it is 0.346574 with standard deviation 0.000000 — deterministic, as Definition 10.1 says it must be. [experimentally confirmed; `mq4_r107`, `delta_r109`]

Remark 10.11 ($\theta = \frac{1}{4}$ is the limit of the maxima, not a maximum). A first scan of $F_A(\theta)^{1/b}$ over a fine net asked whether anything exceeds $1/\sqrt{2}$, and found that something does. That is not a counterexample and the reason is worth stating, because it fixes the form of Theorem 10.8: the maximum is *approached* at $\frac{1}{4}$, and the statement is about the limit of $\max_\theta F_A^{1/b}$, not about where a maximum sits at finite b. Moving off $\frac{1}{4}$ by δ changes $\log F_A^{1/b}$ by $\delta\, b^{-1} \sum_a (\pm \pi a) + O(\delta^2)$, and the signs depend on a mod 4, so the first-order term is a random walk of size $b^{-1}\sqrt{S_2}$. Optimising δ gives an excess of $O(1/b)$ in the exponent — a *bounded multiplicative factor* on F_A, not a geometric one. Measured over $b = 111, 190, 310, 542$: $\max_\theta F_A^{1/b} - 1/\sqrt{2}$ is $9.0 \cdot 10^{-4}$, $1.8 \cdot 10^{-3}$, $1.1 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and $b \log(\sqrt{2} \max_\theta F_A^{1/b})$ stays in $[0.05, 0.89]$ — $O(1)$, as derived. The argmax sits at $\theta = 0.24996, \ldots, 0.25001$. [experimentally confirmed; `mq4_r107`]

11 Two questions from the transfer-function strand, settled

(i) The c_d fold-in is settled, and it goes the transfer-function strand's way. The share of the residual that c_d carries falls like $k^{-2\alpha}$ — measured exponents 1.98, 2.35, 2.99 against predicted 2.00, ≈2.4, 3.00 on the odd numbers, the primes and $\alpha = \frac{3}{2}$, with the squares already below the numerical floor. So Φ as printed is right for every larger k and c_d belongs to E_k permanently; the transfer-function strand's 5.22 now says so. [experimentally confirmed, four profiles, $k \leq 300$; `e8_r103`]

(ii) $c_A = c'_A$ is still open, but sharper. Carried to $k = 300$ the ratio reaches 1.0064 (odd numbers), 1.0001 (primes), 0.9984 ($\alpha = \frac{3}{2}$), and the gap closes like $k^{-2.4}$ — faster than the $1/k$ law both constants sit on. [experimentally confirmed; `e8_r103`] [**TODO** the mechanism is still missing, and a measured rate of convergence is evidence about a limit, not a proof of one. F45 stands: substituting s for λ makes the residual three times worse.]

12 What is missing, stated once

Three statements of this Part are conditional, and they are conditional on the *same* thing. Rather than attach a differently worded caveat to each, we state the missing ingredient once, here, and let each consumer point at it.

Problem 12.1 (the shared missing ingredient). Write out the Edgeworth expansion for the tilted local limit theorem of Proposition 5.20 in region R1, uniformly over the window

$|m - \mu_d(s)| \le W(k)$, with explicit constants at the cumulant budget

$$\varepsilon = \frac{|K_d'''|}{(K_d'')^{3/2}} + \frac{K_d^{(4)}}{(K_d'')^2} + \frac{W(k)\,|K_d'''|}{(K_d'')^2}.$$

This is a classical local-limit computation under a product measure. It is given in §5.3 only to the level of its cumulant budget; what is absent is the explicit constants, not the method.

[open. Regions R2 and R3 of Proposition 5.20 are proved — they quote Lemma 5.8 and Lemma 5.10 — so R1 is the whole of what is missing.]

Who is waiting on it. Exactly three statements, and closing Problem 12.1 completes all three at once:

- **Proposition 5.20** becomes unconditional; R2 and R3 are already proved.
- **Theorem 4.1** (rigidity of the gap series) becomes a theorem. Nothing else is missing from it.
- **Theorem 5.1** (the transfer function) becomes a theorem. Nothing else is missing from it.

Each of the three may therefore be read as *conditional on Problem 12.1 and on nothing else*, and that is the exact sense in which they are stated. No other statement of this Part depends on R1: the coset identity, the biased invariant, the modulus-four theorem and the random rate are independent of it, as is every entry of §14 not listed above.

13 Where this sits, and the algorithmic reading

13.1 Iterating the coarse-graining

Lemma 3.1 is used once, in one direction, and that is not the only thing it says. Written as an operator, $(\mathcal{C}_v f)(t) = v^{-1}\sum_{k<v} f(t+k/v)$ acts on X by $\mathcal{C}_v X = (1-1/v)\log 2 + v^{-1}X(v\,\cdot + \tau_v)$, so X spans a one-dimensional affine invariant subspace of every $\mathcal{C}_v$ at once, and the constants compose: $\mathcal{C}_{v'}\mathcal{C}_v = \mathcal{C}_{vv'}$ on this subspace, with released constants adding as $(1-1/v)+v^{-1}(1-1/v') = 1-1/(vv')$. Three questions we have not attempted, listed so that the omission is deliberate rather than silent. *(i)* Which energies other than X have this property — is $-\log|\cos|$ characterised by being an exact fixed point of the family $\{\mathcal{C}_v\}$ up to affine terms? *(ii)* Is there a transfer-operator reading in which Φ of the transfer-function strand is the spectral object and $\mathcal{C}_v$ the renormalisation map? *(iii)* Parts I and II and the transfer-function strand's landscape language was a source of questions; here it produced an identity. Does the same happen for the *biased* energy of §9, where the coset structure is weighted by q? [conjectural directions; nothing in this paper depends on any of them]

13.2 Related work

Subset sums and number partitioning have a large landscape literature, and it is worth saying precisely where the present series does and does not meet it. That literature is overwhelmingly about *random instances*: the phase transition and barrier-tree statistics of the partitioning landscape, the local-REM description of the near-optimal spectrum, and more recently overlap-gap obstructions for local algorithms. Its objects are the energy spectrum and the dynamics; its randomness is in the instance; and its answers are laws for a typical instance.

Parts I and II and the transfer-function strand do something different in kind: they fix the sequence and compute an *arithmetic invariant of that set*, $\Gamma(A) = 1 + 2\sum_d 2^{-\#\{a\in A:\, a\le 2d\}}$, which the ratio lm/r converges to for *every* target. Nothing is averaged over instances, and the answer is a number one reads off A. The present paper's §10 is the one place where the two meet — there the sequence is random — and even there the question is the ratio's limit rather than the spectrum's law.

One vocabulary warning, since the words collide. "Biased Bernoulli measure" is established terminology in the theory of Bernoulli convolutions, where it denotes a self-similar measure on $\mathbb{R}$; that is a different object from P_q here, and we do not claim the term. Within the subset-sum landscape literature we have not found the biased deformation $\Gamma^{(q)}$ of §8, nor the odd-residue extremal quantity M_{odd} of §10.1, under any name. [literature pass, not a proof of novelty: absence of evidence over the searches made]

13.3 The algorithmic reading

This programme began with local search on subset-sum instances, and the invariant answers a question one asks there. If a local-search procedure ends at a uniformly chosen strict local minimum, the expected number of attempts before landing on a ground state is lm/r, and Parts I and II and the transfer-function strand predict that this number converges to $\Gamma(A)$ — *computable from A alone, without solving a single instance, and independent of the target.* For the odd numbers it is 3; for the primes, 5.3492… Two conditions are attached to that sentence and we attach them here rather than leaving them in §5: the convergence is a *proof skeleton* there, not a theorem (Theorem 4.1, whose missing ingredient is named in its own statement), and the uniformity of the terminal distribution is an assumption about the search, not a fact about the landscape.

Theorem 8.5 is the part of this subsection that has a proof, and it is a statement about *sampling*, not about search dynamics: drawing configurations with a biased coin instead of a fair one multiplies the ratio by $\Gamma^{(q)}/\Gamma \ge 1$, strictly for $q \ne \frac{1}{2}$. So a natural one-parameter family of sampling rules cannot beat the fair coin on this statistic — which is a claim about the statistic, and becomes a claim about a restart heuristic only under the uniformity assumption just named. Section 9 adds that the biased landscape also acquires a *first-order* sensitivity to the target that the fair one does not have.

Behind the uniformity assumption is the reason this is an outlook and not a theorem about algorithms: a count of local minima is not a distribution over basins of attraction, and if the basins are very unequal the ratio bounds the wrong quantity. Nor does anything here address dynamics — barrier heights, escape times, or the behaviour of any particular heuristic. What the invariant supplies is a target-independent constant that any such analysis has to be consistent with, together with a proved statement that one obvious way of trying to improve that constant does not improve it. [**TODO** if this section is ever to make an algorithmic claim, the missing object is the basin-size distribution, not more landscape statistics.]

14 Honest scope

14.1 The deformed and random strand

- *Lean-verified*: Lemma 8.3 in all three forms (Remark 8.4), including the equality case; and the maximisation step of Theorem 10.5 (Remark 10.6), the closed form being taken as a definition there; and the two peak facts of Remark 10.7 (`Pnp/Theory/OddPeaks.lean`); and the odd branch of Lemma 10.3 itself, by the double-angle route (`Pnp/Theory/OddProd.lean`, Remark 10.4). The canon also passes an independent kernel replay (`lean4checker` replays every constant of the import closure of the root module `Pnp` from the imports up, with a negative control). Each file named above is inside that closure; a file present in the source tree but outside it is *not* replayed, and the harness now fails if one exists, so this sentence cannot quietly become false.

- *Proved*: Lemma 8.3, Theorem 8.5 and Corollary 8.7; Lemma 3.1 (twice) with Corollaries 3.2 and 10.2; Theorem 10.5 and Lemma 10.3; Lemma 10.9; and Theorem 10.8, all constants written out. Also Remark 8.2's derivation of Definition 8.1, modulo the biased flatness hypothesis, which is not yet stated.

- *Experimentally confirmed, with the range stated*: the point prediction (§8.2, two profiles, five q, three k), the stratum assignment (Remark 8.8), the linear response under bias (§8.3).

- *Derived and verified*: $\Phi^{(q)}$ of (4) and Proposition 9.1 — three exact internal checks, and the first-order point prediction confirmed against the naive alternative on 36 discriminating rows (§9.4); the random-ensemble floor of §10; the c_d verdict of §11.

- *Interpretation, with nothing deduced from it*: the renormalisation reading of Lemma 3.1 (Remark 3.4) and the three questions of §13.

- *Not yet written*: the passage from (H_q) to a rigidity statement.

14.2 The transfer-function strand

In the style of Part I [3] §7.3 and Part II [4] §11, here is what rests on what.

- *Formally verified*: the combinatorial layer, inherited from Part I, for every target.

- *Proved*: the bridge of §5.2 — Lemma 5.8 in full, Lemma 5.10 in all three parts (with (c) quoted verbatim from Part II's Step 2 at $n = 1$, and the covering apparatus of that paper not needed here, see Remark 5.16), Lemma 5.17, and Proposition 7.1. The minor-arc input is m-independent, so it transfers to off-centre targets unchanged; Part II's ineffectivity through the Siegel–Walfisz constant is inherited at the single point recorded in Remark 5.13.

- *Proof skeleton, with the analytic ingredients in place*: Theorems 4.1 and 5.1. Regions R2 and R3 of Proposition 5.20 are proved — they quote the bridge — and region R1 is the classical local limit computation with p_a-weights, given here to the level of its cumulant budget rather than with every Edgeworth constant written out. That is the one gap between this paper and a complete proof, and it is a gap of exposition rather than of ingredients. Remark 5.23 records that those constants cannot be imported from the effective local-limit literature, and why: our summands have Bernoulli part exactly 0, and their weights make the variance grow like the cube of the number of summands, which is outside the regime that literature is calibrated to.

- *Experimentally confirmed, with the range stated*: rigidity (§4), the transfer function to 0.8–1.9% of the signal (Table 1), universality (§6), the profile-dependent constant $4(2\alpha+1)/(\alpha+1)$ verified on four profiles.

- *Not claimed*: the residual 0.8–1.9% of Table 1 is consistent with finite size and shrinks in k, but we do not claim it is zero.

Remark 14.1 (the single ineffective constant). Precisely: every estimate used here is effective except part (b) of Lemma 5.10, whose error term is the Siegel–Walfisz error, and Part II's appendix identifies the same constant C_1 as the only ineffective one in that paper, entering there at one point as well. Part I is unconditional and effective throughout. **Ineffective is not conditional**: no statement here depends on an unproved hypothesis; what is not available is a numerical threshold beyond which the asymptotics take hold.

Remark 14.2 (approaches that did not work). As in Part II we keep the dead ends, with reasons. *The* Γ_x *family*: the idea that off-centre targets are described by a family of tilted gap series indexed by x. Rejected by the data — rewriting the observed collapse in the exact tilt parameter makes it worse, not better. *The multiplicative bridge*: $|\widetilde{G}_B| \le |G_B|^\kappa$, which would have carried every exponent of Part II across at once. Impossible for the reason in Remark 5.9, and the underlying inequality runs the other way. *The residual as a tilt-versus-slope artefact*: the constants of Remarks 2.5 and 5.7 agree to two digits on every profile, but substituting s for λ makes the residual three times worse — a shared driver, not a cause. *A half-integer exponent* reported in an early draft, which was an artefact of the Gaussian slope (Remark 2.3). *Restating* (H) *so that the cubes violate it*: the series is bounded for the cubes and unbounded for $\alpha < 1$, and the hypothesis follows the computation rather than the narrative.

Use of AI tools

This work was carried out by the author working with AI language models (Anthropic's Claude) as tools, under the author's direction. They were used to fill in Lean proofs, to run and tabulate the numerical experiments, and to draft sections which the author then checked. The choice of problem, the design decisions and the responsibility for every claim here are the author's.

Work produced this way cannot reasonably be taken on the author's word, and the apparatus described under Data availability exists for that reason: every settled theorem is machine-checked in Lean and replayed through the kernel by an independent checker that must first reject three deliberately corrupted modules; every number quoted in the text is produced by a committed script that writes a log beside itself; every theorem, proposition and lemma declares its status where it is stated; and a suite of mechanical checks enforces all of this before each commit. The repository records, in full, the mistakes this process has made.

Data availability

Every number in this paper is produced by a script that writes a log beside itself, as in Parts I and II and the transfer-function strand. The scripts, the logs, the Lean development and the checks that enforce all of this are public at `https://github.com/mathlab0911/arithmetic-landscapes`.

References

[1] OpenAI, *Ten advances in mathematics and theoretical computer science*, 2026, with Lean 4 certificates at `github.com/openai/ten-proofs`. Cited here only as a precedent for the genre; no mathematical result of that work is used.

[2] R. Giuliano and M. Weber, *Approximate local limit theorems with effective rate and application to random walks in random scenery*. Bernoulli **23** (2017), 3268–3310; arXiv:1412.3980. Cited only for the definition and the applicability condition quoted in Remark 5.23.

[3] K. Amauchi, *Arithmetic landscapes I: the gap series*. Manuscript, 2026.

[4] K. Amauchi, *Arithmetic landscapes II: asymptotic flatness of subset-sum landscapes of primes*. Manuscript, 2026.

[5] K. Amauchi, *The transfer function of subset-sum landscapes*. Manuscript, 2026. Being absorbed into the present Part; see the note on scope.